Appendices for "Computer Modeling of Mitochondrial TCA Cycle, Oxidative Phosphorylation, Metabolite Transport, and Electrophysiology"

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Appendix A: List of Model Components

This appendix lists the basic components of the model. Tables A1 and A2 list the state variables and reaction and transport fluxes considered in the model. Tables A3 and A4 list the thermodynamic data and fixed model parameters.

Table A1: Model Variables

Variables	Descriptions	Units
$\Delta\Psi$	Mitochondrial membrane potential	mV
Mitochondrial Matrix Variables:		
$[H^+]_x$	Concentration of H^+ ion in mito matrix	mol (l matrix water)$^{-1}$
$[K^+]_x$	Concentration of K^+ ion in mito matrix	mol (l matrix water)$^{-1}$
$[Mg^{2+}]_x$	Concentration of Mg^{2+} ion in mito matrix	mol (l matrix water)$^{-1}$
$[NADH]_x$	Concentration of NADH in mito matrix	mol (l matrix water)$^{-1}$
$[NAD]_x$	Concentration of NAD in mito matrix	mol (l matrix water)$^{-1}$
$[QH_2]_x$	Concentration of reduced ubiquinol in mito matrix	mol (l matrix water)$^{-1}$
$[COQ]_x$	Concentration of oxidized ubiquinol in mito matrix	mol (l matrix water)$^{-1}$
$[ATP]_x$	Concentration of total ATP in mito matrix	mol (l matrix water)$^{-1}$
$[ADP]_x$	Concentration of total ADP in mito matrix	mol (l matrix water)$^{-1}$
$[GTP]_x$	Concentration of total GTP in mito matrix	mol (l matrix water)$^{-1}$
$[GDP]_x$	Concentration of total GDP in mito matrix	mol (l matrix water)$^{-1}$
$[PI]_x$	Concentration of inorganic phosphate in mito matrix	mol (l matrix water)$^{-1}$
$[PYR]_x$	Concentration of pyruvate in mito matrix	mol (l matrix water)$^{-1}$
$[COASH]_x$	Concentration of CoA-SH in mito matrix	mol (l matrix water)$^{-1}$
$[ACCOA]_x$	Concentration of acetyl-CoA in mito matrix	mol (l matrix water)$^{-1}$
$[OAA]_x$	Concentration of oxaloacetate in mito matrix	mol (l matrix water)$^{-1}$
$[CIT]_x$	Concentration of citrate in mito matrix	mol (l matrix water)$^{-1}$
$[ICIT]_x$	Concentration of isocitrate in mito matrix	mol (l matrix water)$^{-1}$
$[AKG]_x$	Concentration of α-ketoglutarate in mito matrix	mol (l matrix water)$^{-1}$

$[SCOA]_x$	Concentration of succinyl-CoA in mito matrix	mol (l matrix water) $^{-1}$
$[SUC]_x$	Concentration of pyruvate in mito matrix	mol (l matrix water) $^{-1}$
$[FUM]_x$	Concentration of fumarate in mito matrix	mol (l matrix water) $^{-1}$
$[MAL]_x$	Concentration of malate in mito matrix	mol (l matrix water) $^{-1}$
$[ASP]_x$	Concentration of aspartate in mito matrix	mol (l matrix water) $^{-1}$
$[GLU]_x$	Concentration of glutamate in mito matrix	mol (l matrix water) $^{-1}$
$[O_2]_x$	Concentration of oxygen in mito matrix	mol (l matrix water) $^{-1}$
$[CO_2tot]_x$	Concentration of total CO_2 in mito matrix	mol (l matrix water) $^{-1}$
Mitochondrial Inter-Membrane Space Variables:		
$[H^+]_i$	Concentration of H^+ ion in IM space	mol (l IM water) $^{-1}$
$[K^+]_i$	Concentration of K^+ ion in IM space	mol (l IM water) $^{-1}$
$[Mg^{2+}]_i$	Concentration of Mg^{2+} ion in IM space	mol (l IM water) $^{-1}$
$[Cred]_i$	Concentration of reduced cytochrome C in IM space	mol (l IM water) $^{-1}$
$[Cox]_i$	Concentration of oxidized cytochrome C in IM space	mol (l IM water) $^{-1}$
$[ATP]_i$	Concentration of total ATP in IM space	mol (l IM water) $^{-1}$
$[ADP]_i$	Concentration of total ADP in IM space	mol (l IM water) $^{-1}$
$[AMP]_i$	Concentration of total AMP in IM space	mol (l IM water) $^{-1}$
$[PI]_i$	Concentration of inorganic phosphate in IM space	mol (l IM water) $^{-1}$
$[PYR]_i$	Concentration of pyruvate in IM space	mol (l IM water) $^{-1}$
$[CIT]_i$	Concentration of citrate in IM space	mol (l IM water) $^{-1}$
$[AKG]_i$	Concentration of α-ketoglutarate in IM space	mol (l IM water) $^{-1}$
$[SUC]_i$	Concentration of pyruvate in IM space	mol (l IM water) $^{-1}$
$[FUM]_i$	Concentration of fumarate in IM space	mol (l IM water) $^{-1}$
$[MAL]_i$	Concentration of malate in IM space	mol (l IM water) $^{-1}$
$[ASP]_i$	Concentration of ssparatate in IM space	mol (l IM water) $^{-1}$
$[GLU]_i$	Concentration of glutamate in IM space	mol (l IM water) $^{-1}$
External Space Variables:		
$[H^+]_c$	Concentration of H^+ ion in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[K^+]_c$	Concentration of K^+ ion in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[Mg^{2+}]_c$	Concentration of Mg^{2+} ion in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[ATP]_c$	Concentration of total ATP in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[ADP]_c$	Concentration of total ADP in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[AMP]_c$	Concentration of total ADP in buffer/cyto	mol (l buffer/cyto water) $^{-1}$
$[PI]_c$	Concentration of inorganic phosphate in buffer/cyto	mol (l buffer/cyto water) $^{-1}$

$[PYR]_c$	Concentration of pyruvate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[CIT]_c$	Concentration of citrate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[AKG]_c$	Concentration of α-ketoglutarate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[SUC]_c$	Concentration of pyruvate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[FUM]_c$	Concentration of fumarate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[MAL]_c$	Concentration of malate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[ASP]_c$	Concentration of aspartate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[GLU]_c$	Concentration of glutamate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[GLC]_c$	Concentration of glucose in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[G6P]_c$	Concentration of glucose-6-phosphate in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[PCr]_c$	Concentration of phosphate creatine in buffer/cyto	mol (l buffer/cyto water)$^{-1}$
$[Cr]_c$	Concentration of creatine in buffer/cyto	mol (l buffer/cyto water)$^{-1}$

Table A2: Reaction and Transport Fluxes

Flux	Description	Units
Mitochondrial Reactions:		
J_{C1}	Complex I	mol s^{-1} (l mito)$^{-1}$
J_{C3}	Complex III	mol s^{-1} (l mito)$^{-1}$
J_{C4}	Complex IV	mol s^{-1} (l mito)$^{-1}$
J_{F1}	F_1F_0 ATPase reaction	mol s^{-1} (l mito)$^{-1}$
J_{ANT}	Adenine nucleotide translocase	mol s^{-1} (l mito)$^{-1}$
J_{PIHt}	Phosphate-hydrogen co-transporter	mol s^{-1} (l mito)$^{-1}$
J_{Hle}	Proton leak	mol s^{-1} (l mito)$^{-1}$
J_{KH}	Mitochondrial K^+ / H^+ exchanger	mol s^{-1} (l mito)$^{-1}$
J_{pdh}	Pyruvate dehydrogenase	mol s^{-1} (l mito)$^{-1}$
J_{cits}	Citrate synthetase	mol s^{-1} (l mito)$^{-1}$
J_{acon}	Aconitase	mol s^{-1} (l mito)$^{-1}$
J_{isod}	Isocitrate dehydrogenase	mol s^{-1} (l mito)$^{-1}$
J_{akgd}	α-Ketoglutarate dehydrogenase	mol s^{-1} (l mito)$^{-1}$
J_{scoas}	Succinyl-CoA synthetase	mol s^{-1} (l mito)$^{-1}$
J_{sdh}	Succinate dehydrogenase	mol s^{-1} (l mito)$^{-1}$
J_{fum}	Fumarase	mol s^{-1} (l mito)$^{-1}$
J_{mdh}	Malate dehydrogenase	mol s^{-1} (l mito)$^{-1}$
J_{ndk}	Nucleoside diphosphokinase	mol s^{-1} (l mito)$^{-1}$

J_{got}	Glutamate oxaloacetate transaminase (Aspartate Transaminase)	mol s^{-1} (l mito)$^{-1}$
J_{AKi}	Mitochondrial adenylate kinase	mol s^{-1} (l mito)$^{-1}$
Mitochondrial Transport Fluxes:		
J_{PYRH}	Pyruvate-H^+ co-transporter	mol s^{-1} (l mito)$^{-1}$
J_{GLUH}	Glutamate-H^+ co-transporter	mol s^{-1} (l mito)$^{-1}$
J_{CITMAL}	Citrate/malate antiporter	mol s^{-1} (l mito)$^{-1}$
J_{AKGMAL}	α-Ketoglutarate/malate antiporter	mol s^{-1} (l mito)$^{-1}$
J_{SUCMAL}	Succinate/malate antiporter	mol s^{-1} (l mito)$^{-1}$
J_{MALPI}	Malate/phosphate antiporter	mol s^{-1} (l mito)$^{-1}$
J_{ASPGLU}	Aspartate/glutamate antiporter	mol s^{-1} (l mito)$^{-1}$
J_{PIt}	Phosphate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{ATPt}	ATP transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{ADPt}	ADP transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{AMPt}	AMP transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{PYRt}	Pyruvate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{CITt}	Citrate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{MALt}	Malate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{AKGt}	α-Ketoglutarate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{SUCt}	Succinate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{GLUt}	Glutamate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
J_{ASPt}	Aspartate transport across outer membrane	mol s^{-1} (l mito)$^{-1}$
Buffer/Cytoplasm Space Reactions:		
J_{HK}	Hexokinase	mol s^{-1} (l buffer)$^{-1}$
J_{AtC}	ATP hydrolysis	mol s^{-1} (l cyto)$^{-1}$
J_{AKc}	Cytoplasmic adenylate kinase	mol s^{-1} (l cyto)$^{-1}$
J_{CK}	Creatine kinase	mol s^{-1} (l cyto)$^{-1}$

Table A3: List of Reactant, Abbreviation, Reference Species, Gibbs Free Energy of Formation, and Binding Constants (298.15 K, 1 M reactants, I = 0.17 M, P = 1 atm)

Reactant	Reference species	$\Delta_f G_o$ (kJ/mol)	Ion-bound species	pK
H_2O	H_2O	-235.74	-	-

O_2 (aq)	O_2 (aq)	16.40	-	-
NADH	$NADH^0$	39.31	-	-
NAD	NAD^+	18.10	-	-
QH2	QH_2^0	-23.30	-	-
COQ	COQ	65.17	-	-
ATP	ATP^{4-}	-2771.00	$HATP^{3-}$	6.59
			$MgATP^2$	3.82[a]
			$KATP^{3-}$	1.87[a]
ADP	ADP^{3-}	-1903.96	$HADP^{2-}$	6.42
			$MgADP^-$	2.79[a]
			$KADP^{2-}$	1.53[a]
AMP	AMP^{2-}	-1034.66	$HAMP^-$	6.22
			$MgAMP^0$	1.86[a]
			$KAMP^-$	1.05[a]
Cred	$Cred^{2+}$	-27.41	-	-
Cox	Cox^{3+}	-6.52	-	-
PI	HPO_4^{2-}	-1098.27	$H_2PO_4^-$	6.71
			$MgHPO_4^0$	1.69[a]
			$KHPO_4^-$	0.0074[a]
PCr	PCr^{2-}	-	-	-
Cr	HCr^0	-252.68	-	-
$FADH_2$	$FADH_2^0$	-67.60	-	-
FAD	FAD^0	19.55	-	-
COASH	$COAS^-$	-0.72	$COASH^0$	8.13
ACCOA	$ACCOA^0$	-178.19	-	-
OAA	OAA^{2-}	-794.41	$MgOAA^0$	0.0051[a]
CIT	CIT^{3-}	-1165.59	$HCIT^{2-}$	5.63
			$MgCIT^-$	3.37[a]
			$KCIT^{2-}$	0.339[a]
ICIT	$ICIT^{3-}$	-1158.94	$HICIT^{2-}$	5.64
			$MgICIT^-$	2.46[a]
AKG	AKG^{2-}	-793.41	-	-
SCOA	$SCOA^-$	-507.55	$HSCOA^0$	3.96
SUC	SUC^{2-}	-690.44	$HSUC^-$	5.13

			$MgSUC^0$	1.17[a]
			$KSUC^-$	0.503[a]
FUM	FUM^{2-}	-603.32	$HFUM^-$	4.10
MAL	MAL^{2-}	-842.66	$HMAL^-$	4.75
			$MgMAL^0$	1.55[a]
			$KMAL^-$	-0.107[a]
ASP	ASP^-	-692.26	$HASP^0$	3.65[a]
			$MgASP^+$	2.32[a]
GLU	GLU^-	-692.40	$HGLU^0$	4.06[a]
			$MgGLU^+$	1.82[a]
PYR	PYR^-	-470.82	$MgPYR^+$	0.962[a]
GLC	GLC^0	-907.21	-	-
G6P	$G6P^{2-}$	-1758.87	$HG6P^-$	5.85
CO_2tot	CO_2^{3-}	-530.71	HCO_3^-	9.75

All values from (2) unless otherwise noted.
[a] NIST database 46: Critical Stability Constants

Table A4: Fixed Model Parameter

Parameter	Description	Value	Units	Reference
Physicochemical Parameters:				
RT	Gas constant times temperature	2.5775	kJ mol^{-1}	–[a]
F	Faraday's constant	0.096484	kJ mol^{-1} mV^{-1}	–[a]
Structure/Volume Parameters:				
V_{cyto}	Cytoplasm Volume	0.894	(l cytoplasm) $(\text{l cell})^{-1}$	(55)
V_{mito}	Mitochondrial Volume	0.056	(l mito) $(\text{l cell})^{-1}$	(53)
W_x	Matrix water space fraction	0.6514	(l water) $(\text{l mito})^{-1}$	(4, 52)
W_i	IM space water fraction	0.0724	(l water) $(\text{l mito})^{-1}$	(4, 52)
W_c	Buffer water fraction	80	(l buffer water) $(\text{l mito})^{-1}$	(30)[b]
	Cytoplasm water fraction	0.8425	(l water) $(\text{l cyto})^{-1}$	(52)
γ	Outer membrane area per mito volume	5.99	μm^{-1}	(34)
ρ_m	Protein density of mitochondria	2.725×10^5	(mg Protein) $(\text{l mito})^{-1}$	(52)
Mitochondrial Model Parameters:				
n_A	H^+ stoich. coef. for F_1F_0-ATPase	3	unitless	(47)

p_{PI}	Mitochondrial membrane permeability to inorganic phosphate	327	μm sec^{-1}	(51)
p_A	Mitochondrial outer membrane permeability to nucleotides	85.0	μm sec^{-1}	(31)
$k_{m,ADP}$	ANT Michaelis-Menten constant	3.5×10^{-6}	M	(51)[d]
θ	ANT parameter	0.60	unitless	(51)[b]
k_{O2}	Kinetic constant for complex IV	1.2×10^{-4}	M	(51)[d]
β	Matrix buffering capacity	0.01	M	(51)[d]
C_{IM}	Capacitance of inner membrane	6.75×10^{-6}	mol (l mito) $^{-1}$ mV^{-1}	(4, 14)
B_x	Matrix buffering parameter	0.02	M	(21, 49)[b]
K_{Bx}	Matrix buffering parameter	1×10^{-7}	M	(21, 49)[b]
Fixed Concentrations and Concentration Pools:				
NAD_{tot}	Total matrix NAD(H) concentration	2.97	mol (l matrix water) $^{-1}$	(51)[c]
Q_{tot}	Total matrix ubiquinol concentration	1.35	mol (l matrix water) $^{-1}$	(51)[c]
$cytC_{tot}$	Total IM cytochrome c concentration	2.70	mol (l IM water) $^{-1}$	(51)[c]
A_{tot}	Total matrix ATP+ADP concentration	10	mol (l matrix water) $^{-1}$	(51)[c]
CR_{tot}	Total Cr+CrP concentration	42.7	mol (l cytoplasm water) $^{-1}$	(20)
$[CO_2tot]_x$	Total CO_2 concentration in the matrix	21.4×10^{-3}	Molar	(50)
Other Constants:				
K_{CK}	Creatine kinase equilibrium constant	3.57×10^{8}	M^{-1}	(2)[d]

[a]Standard physicochemical constants.
[b]Values adjusted to match experimental data
[c]Value used is taken from previous modeling studies, not direct experimental measure.
[d]Value used is modified to match thermodynamic data.

Appendix B: Computational Model

The model is mathematically described by using the differential equations listed in this appendix. The oxidative phosphorylation component of the model is derived from previously published work (4, 5, 55). The details behind the TCA cycle enzyme kinetic schemes are provided in Appendix C. Here the subscripts “x”, “i”, and “c” on variable names denote matrix, intermembrane, and cytoplasmic (extra-mitochondrial) spaces, respectively. For example $[\mathrm{ATP}]_\mathrm{x}$ denotes matrix ATP concentration while $[\mathrm{ATP}]_\mathrm{c}$ denotes ATP concentration in the cytoplasm or buffer space for an isolated mitochondria experiment.

B1. Differential Equations

The differential equations are grouped into equations for membrane potential, mitochondrial matrix variables, intermembrane space variables, and external space variables. The external space variables used depend on the type of simulation considered (ex vivo isolated mitochondrial or the in vivo system.)

Mitochondrial Inner Membrane Electrical Potential:

$$d\Delta\Psi / dt = (+4J_{\mathrm{C1}} + 2J_{\mathrm{C3}} + 4J_{\mathrm{C4}} - n_{\mathrm{A}} J_{\mathrm{F1}} - J_{\mathrm{ANT}} - J_{\mathrm{Hle}} + J_{\mathrm{ASPGLU}}) / C_{\mathrm{IM}} \qquad \text{(B1)}$$

Mitochondrial Matrix:

$$d[\mathrm{ATP}]_\mathrm{x} / dt = (+J_{\mathrm{ndk}} + J_{\mathrm{F1}} - J_{\mathrm{ANT}}) / W_\mathrm{x} \qquad \text{(B2)}$$

$$d[\mathrm{ADP}]_\mathrm{x} / dt = (-J_{\mathrm{ndk}} - J_{\mathrm{F1}} + J_{\mathrm{ANT}}) / W_\mathrm{x} \qquad \text{(B3)}$$

$$d[\mathrm{AMP}]_\mathrm{x} / dt = 0 / W_\mathrm{x} \qquad \text{(B4)}$$

$$d[\mathrm{GTP}]_\mathrm{x} / dt = (+J_{\mathrm{scoas}} - J_{\mathrm{ndk}}) / W_\mathrm{x} \qquad \text{(B5)}$$

$$d[\mathrm{GDP}]_\mathrm{x} / dt = (-J_{\mathrm{scoas}} + J_{\mathrm{ndk}}) / W_\mathrm{x} \qquad \text{(B6)}$$

$$d[\mathrm{PI}]_\mathrm{x} / dt = (-J_{\mathrm{scoas}} - J_{\mathrm{F1}} + J_{\mathrm{PIHt}} - J_{\mathrm{SUCPI}} - J_{\mathrm{MALPI}}) / W_\mathrm{x} \qquad \text{(B7)}$$

$$d[\mathrm{NADH}]_\mathrm{x} / dt = (+J_{\mathrm{pdh}} + J_{\mathrm{isod}} + J_{\mathrm{akgd}} + J_{\mathrm{mdh}} - J_{\mathrm{C1}}) / W_\mathrm{x} \qquad \text{(B8)}$$

$$d[\mathrm{QH_2}]_\mathrm{x} / dt = (+J_{\mathrm{sdh}} + J_{\mathrm{C1}} - J_{\mathrm{C3}}) / W_\mathrm{x} \qquad \text{(B9)}$$

$$d[\mathrm{PYR}]_\mathrm{x} / dt = (-J_{\mathrm{pdh}} + J_{\mathrm{PYRH}}) / W_\mathrm{x} \qquad \text{(B10)}$$

$$d[\mathrm{ACCOA}]_\mathrm{x} / dt = (-J_{\mathrm{cits}} + J_{\mathrm{pdh}}) / W_\mathrm{x} \qquad \text{(B11)}$$

$$d[\mathrm{CIT}]_\mathrm{x} / dt = (+J_{\mathrm{cits}} - J_{\mathrm{acon}} + J_{\mathrm{CITMAL}}) / W_\mathrm{x} \qquad \text{(B12)}$$

$$d[\mathrm{ICIT}]_\mathrm{x} / dt = (+J_{\mathrm{acon}} - J_{\mathrm{isod}}) / W_\mathrm{x} \qquad \text{(B13)}$$

$$d[\mathrm{AKG}]_\mathrm{x} dt = (+J_{\mathrm{isod}} - J_{\mathrm{akgd}} - J_{\mathrm{got}} + J_{\mathrm{AKGMAL}}) / W_\mathrm{x} \qquad \text{(B14)}$$

$$d[\mathrm{SCOA}]_\mathrm{x} / dt = (+J_{\mathrm{akgd}} - J_{\mathrm{scoas}}) / W_\mathrm{x} \qquad \text{(B15)}$$

$$d[\mathrm{COASH}]_\mathrm{x} / dt = (-J_{\mathrm{pdh}} - J_{\mathrm{akgd}} + J_{\mathrm{scoas}} + J_{\mathrm{cits}}) / W_\mathrm{x} \qquad \text{(B16)}$$

$$d[\mathrm{SUC}]_\mathrm{x} / dt = (+J_{\mathrm{scoas}} - J_{\mathrm{sdh}} + J_{\mathrm{SUCMAL}}) / W_\mathrm{x} \qquad \text{(B17)}$$

$$d[\mathrm{FUM}]_\mathrm{x} / dt = (+J_{\mathrm{sdh}} - J_{\mathrm{fum}}) / W_\mathrm{x} \qquad \text{(B18)}$$

$$d[\mathrm{MAL}]_\mathrm{x} / dt = (+J_{\mathrm{fum}} - J_{\mathrm{mdh}} + J_{\mathrm{MALPI}} - J_{\mathrm{AKGMAL}} - J_{\mathrm{CITMAL}} - J_{\mathrm{SUCMAL}}) / W_\mathrm{x} \qquad \text{(B19)}$$

$$d[\mathrm{OAA}]_\mathrm{x} / dt = (-J_{\mathrm{cits}} + J_{\mathrm{mdh}} + J_{\mathrm{got}}) / W_\mathrm{x} \qquad \text{(B20)}$$

$$d[\mathrm{GLU}]_\mathrm{x} / dt = (+J_\mathrm{got} + J_\mathrm{GLUH} - J_\mathrm{ASPGLU}) / W_\mathrm{x} \quad \text{(B21)}$$

$$d[\mathrm{ASP}]_\mathrm{x} / dt = (-J_\mathrm{got} + J_\mathrm{ASPGLU}) / W_\mathrm{x} \quad \text{(B22)}$$

$$d[\mathrm{O_2}]_\mathrm{x} / dt = 0 / W_\mathrm{x} \quad \text{(B23)}$$

$$d[\mathrm{CO_2tot}]_\mathrm{x} / dt = 0 / W_\mathrm{x}\,. \quad \text{(B24)}$$

Mitochondrial Inter-Membrane Space:

$$d[\mathrm{Cred}]_\mathrm{i} / dt = (+2J_\mathrm{C3} - 2J_\mathrm{C4}) / W_\mathrm{i} \quad \text{(B25)}$$

$$d[\mathrm{ATP}]_\mathrm{i} / dt = (+J_\mathrm{ATP} + J_\mathrm{ANT} + J_\mathrm{AKi}) / W_\mathrm{i} \quad \text{(B26)}$$

$$d[\mathrm{ADP}]_\mathrm{i} / dt = (+J_\mathrm{ADP} - J_\mathrm{ANT} - 2J_\mathrm{AKi}) / W_\mathrm{i} \quad \text{(B27)}$$

$$d[\mathrm{AMP}]_\mathrm{i} / dt = (J_\mathrm{AMPt} + J_\mathrm{AKi}) / W_\mathrm{i} \quad \text{(B28)}$$

$$d[\mathrm{PI}]_\mathrm{i} / dt = (-J_\mathrm{PIHt} + J_\mathrm{PIt} + J_\mathrm{MALPI}) / W_\mathrm{i} \quad \text{(B29)}$$

$$d[\mathrm{PYR}]_\mathrm{i} / dt = (-J_\mathrm{PYRH} + J_\mathrm{PYRt}) / W_\mathrm{i} \quad \text{(B30)}$$

$$d[\mathrm{CIT}]_\mathrm{i} / dt = (-J_\mathrm{CITMAL} + J_\mathrm{CITt}) / W_\mathrm{i} \quad \text{(B31)}$$

$$d[\mathrm{ICIT}]_\mathrm{i} / dt = (-J_\mathrm{ICITMAL} + J_\mathrm{ICITt}) / W_\mathrm{i} \quad \text{(B32)}$$

$$d[\mathrm{AKG}]_\mathrm{i} / dt = (-J_\mathrm{AKGMAL} + J_\mathrm{AKGt}) / W_\mathrm{i} \quad \text{(B33)}$$

$$d[\mathrm{SUC}]_\mathrm{i} / dt = (+J_\mathrm{SUCt} - J_\mathrm{SUCMAL}) / W_\mathrm{i} \quad \text{(B34)}$$

$$d[\mathrm{FUM}]_\mathrm{i} / dt = 0 / W_\mathrm{i} \quad \text{(B35)}$$

$$d[\mathrm{MAL}]_\mathrm{i} / dt = (-J_\mathrm{MALPI} + J_\mathrm{MALt} + J_\mathrm{AKGMAL} + J_\mathrm{CITMAL} + J_\mathrm{SUCMAL}) / W_\mathrm{i} \quad \text{(B36)}$$

$$d[\mathrm{GLU}]_\mathrm{i} / dt = (-J_\mathrm{GLUH} + J_\mathrm{ASPGLU} + J_\mathrm{GLUt}) / W_\mathrm{i} \quad \text{(B37)}$$

$$d[\mathrm{ASP}]_\mathrm{i} / dt = (-J_\mathrm{ASPGLU} + J_\mathrm{ASPt}) / W_\mathrm{i}\,. \quad \text{(B38)}$$

External Space, Ex Vivo Isolated Mitochondria:

$$d[\mathrm{ATP}]_\mathrm{c} / dt = (-J_\mathrm{ATPt} - J_\mathrm{HK}(1 + W_\mathrm{c})) / W_\mathrm{c} \quad \text{(B39)}$$

$$d[\mathrm{ADP}]_\mathrm{c} / dt = (-J_\mathrm{ADPt} + J_\mathrm{HK}(1 + W_\mathrm{c})) / W_\mathrm{c} \quad \text{(B40)}$$

$$d[\mathrm{AMP}]_\mathrm{c} / dt = 0 / W_\mathrm{c} \quad \text{(B41)}$$

$$d[\mathrm{PI}]_\mathrm{c} / dt = -J_\mathrm{PIt} / W_\mathrm{c} \quad \text{(B42)}$$

$$d[\mathrm{PYR}]_\mathrm{c} / dt = -J_\mathrm{PYRt} / W_\mathrm{c} \quad \text{(B43)}$$

$$d[\mathrm{CIT}]_\mathrm{c} / dt = -J_\mathrm{CITt} / W_\mathrm{c} \quad \text{(B44)}$$

$$d[\mathrm{ICIT}]_\mathrm{c} / dt = -J_\mathrm{ICITt} / W_\mathrm{c} \quad \text{(B45)}$$

$$d[\mathrm{AKG}]_\mathrm{c} / dt = -J_\mathrm{AKGt} / W_\mathrm{c} \quad \text{(B46)}$$

$$d[\mathrm{SUC}]_\mathrm{c} / dt = -J_\mathrm{SUCt} / W_\mathrm{c} \quad \text{(B47)}$$

$$d[\mathrm{FUM}]_\mathrm{c} / dt = 0 / W_\mathrm{c} \quad \text{(B48)}$$

$$d[\mathrm{MAL}]_{\mathrm{c}} / dt = -J_{\mathrm{MALt}} / W_{\mathrm{c}} \tag{B49}$$

$$d[\mathrm{GLU}]_{\mathrm{c}} / dt = -J_{\mathrm{GLUt}} / W_{\mathrm{c}} \tag{B50}$$

$$d[\mathrm{ASP}]_{\mathrm{c}} / dt = -J_{\mathrm{ASPt}} / W_{\mathrm{c}} \tag{B51}$$

$$d[\mathrm{GLC}]_{\mathrm{c}} / dt = -J_{\mathrm{HK}} (1 + W_{\mathrm{c}}) / W_{\mathrm{c}} \tag{B52}$$

$$d[\mathrm{G6P}]_{\mathrm{c}} / dt = +J_{\mathrm{HK}} (1 + W_{\mathrm{c}}) / W_{\mathrm{c}} \tag{B53}$$

$$d[\mathrm{PCr}]_{\mathrm{c}} / dt = 0 / W_{\mathrm{c}}\,; \tag{B54}$$

External Space, In Vivo:

$$d[\mathrm{ATP}]_{\mathrm{c}} / dt = (-(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{ATPt}} - J_{\mathrm{AtC}} + J_{\mathrm{CK}} + J_{\mathrm{AKc}}) / W_{\mathrm{c}} \tag{B55}$$

$$d[\mathrm{ADP}]_{\mathrm{c}} / dt = (-(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{ADPt}} + J_{\mathrm{AtC}} - J_{\mathrm{CK}} - 2J_{\mathrm{AKc}}) / W_{\mathrm{c}} \tag{B56}$$

$$d[\mathrm{AMP}]_{\mathrm{c}} / dt = (-(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{AMPt}} + J_{\mathrm{AKc}}) / W_{\mathrm{c}} \tag{B57}$$

$$d[\mathrm{PI}]_{\mathrm{c}} / dt = (-(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{PIt}} + J_{\mathrm{AKc}}) / W_{\mathrm{c}} \tag{B58}$$

$$d[\mathrm{PYR}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{PYRt}} / W_{\mathrm{c}} \tag{B59}$$

$$d[\mathrm{CIT}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{CITt}} / W_{\mathrm{c}} \tag{B60}$$

$$d[\mathrm{AKG}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{AKGt}} / W_{\mathrm{c}} \tag{B61}$$

$$d[\mathrm{SUC}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{SUCt}} / W_{\mathrm{c}} \tag{B62}$$

$$d[\mathrm{FUM}]_{\mathrm{c}} / dt = 0 / W_{\mathrm{c}} \tag{B63}$$

$$d[\mathrm{MAL}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{MALt}} / W_{\mathrm{c}} \tag{B64}$$

$$d[\mathrm{GLU}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{GLUt}} / W_{\mathrm{c}} \tag{B65}$$

$$d[\mathrm{ASP}]_{\mathrm{c}} / dt = -(V_{\mathrm{cyto}} / V_{\mathrm{mito}}) J_{\mathrm{ASPt}} / W_{\mathrm{c}} \tag{B66}$$

$$d[\mathrm{PCr}]_{\mathrm{c}} / dt = -J_{\mathrm{CKc}} / W_{\mathrm{c}}\,. \tag{B67}$$

Assuming constant total concentrations $\mathrm{NAD_{tot}}$, $\mathrm{Q_{tot}}$, $\mathrm{cytC_{tot}}$, and $\mathrm{A_{tot}}$ for nicotinamide nucleotides, ubiquinol, and cytochrome c, we compute concentrations of the following reactants as:

$$[\mathrm{NAD}]_{\mathrm{x}} = \mathrm{NAD_{tot}} - [\mathrm{NADH}]_{\mathrm{x}} \tag{B68}$$

$$[\mathrm{COQ}]_{\mathrm{x}} = \mathrm{Q_{tot}} - [\mathrm{QH_2}]_{\mathrm{x}} \tag{B69}$$

$$[\mathrm{Cox}]_{\mathrm{i}} = \mathrm{cytC_{tot}} - [\mathrm{Cred}]_{\mathrm{i}}\,. \tag{B70}$$

Concentrations of cytoplasmic H^+, Mg^{2+}, and K^+ are assumed to be fixed at buffer conditions or physiological in vivo values. Since the outer membrane is highly permeable to hydrogen ions and cations, we assume here $[\mathrm{H^+}]_{\mathrm{i}} = [\mathrm{H^+}]_{\mathrm{c}}$, $[\mathrm{Mg^{2+}}]_{\mathrm{i}} = [\mathrm{Mg^{2+}}]_{\mathrm{c}}$, and $[\mathrm{K^+}]_{\mathrm{i}} = [\mathrm{K^+}]_{\mathrm{c}}$. Mathematical expressions for time derivatives of [H+], [Mg^{2+}], and [K^+] are presented in Appendix D.

B2. Flux Expressions

Mathematical expressions for oxidative phosphorylation fluxes

Complex I flux:

$$J_{\mathrm{C1}} = X_{\mathrm{C1}}\left(K_{eq,\mathrm{C1}}[\mathrm{NADH}]_{\mathrm{x}}[\mathrm{COQ}]_{\mathrm{x}} - [\mathrm{NAD}]_{\mathrm{x}}[\mathrm{QH_2}]_{\mathrm{x}}\right), \tag{B71}$$

where $K_{eq,\mathrm{C1}} = K^0_{eq,\mathrm{C1}} \cdot [\mathrm{H^+}]^5_{\mathrm{x}} / [\mathrm{H^+}]^4_{\mathrm{i}}$, with $K^0_{eq,C1} = \exp\left(-\left(\Delta_r G^0_{\mathrm{C1}} + 4F\Delta\Psi\right)/RT\right)$ and $\Delta_r G^0_{\mathrm{C1}} = \Delta_f G^0_{\mathrm{NAD}} + \Delta_f G^0_{\mathrm{QH_2}} - \Delta_r G^0_{\mathrm{NADH}} - \Delta_r G^0_{\mathrm{COQ}} = -109.7$ kJ/mol.

Complex III flux:

$$J_{\mathrm{C3}} = X_{\mathrm{C3}}\left(\frac{1+[\mathrm{PI}]_{\mathrm{x}}/k_{\mathrm{PI,1}}}{1+[\mathrm{PI}]_{\mathrm{x}}/k_{\mathrm{PI,2}}}\right)\left(K^{1/2}_{eq,\mathrm{C3}}[\mathrm{Cox}]_{\mathrm{i}}[\mathrm{QH_2}]^{1/2}_{\mathrm{x}} - [\mathrm{Cred}]_{\mathrm{i}}[\mathrm{COQ}]^{1/2}_{\mathrm{x}}\right). \tag{B72}$$

where $K_{eq,\mathrm{C3}} = K^0_{eq,\mathrm{C3}} \cdot [\mathrm{H^+}]^2_{\mathrm{x}} / [\mathrm{H^+}]^4_{\mathrm{i}}$, with $K^0_{eq,C3} = \exp\left(-\left(\Delta_r G^0_{\mathrm{C3}} + 2F\Delta\Psi\right)/RT\right)$ and $\Delta_r G^0_{\mathrm{C3}} = \Delta_f G^0_{\mathrm{Cred}} + \Delta_f G^0_{\mathrm{COQ}} - \Delta_r G^0_{\mathrm{QH2}} - \Delta_r G^0_{\mathrm{Cox}} = 46.69$ kJ/mol.

Complex IV flux:

$$J_{\mathrm{C4}} = X_{\mathrm{C4}}\left(\frac{1}{1+k_{\mathrm{O_2}}/[\mathrm{O_2}]}\right)\exp\left(\frac{F\Delta\Psi}{RT}\right)\left(\frac{[\mathrm{Cred}]_{\mathrm{i}}}{\mathrm{cytC_{tot}}}\right)\left(K^{1/2}_{eq,\mathrm{C4}}[\mathrm{Cred}]_{\mathrm{i}}[\mathrm{O_2}]^{1/4}_{\mathrm{x}} - [\mathrm{Cox}]_{\mathrm{i}}\right). \tag{B73}$$

where $K_{eq,\mathrm{C4}} = K^0_{eq,\mathrm{C4}} \cdot [\mathrm{H^+}]^4_{\mathrm{x}} / [\mathrm{H^+}]^2_{\mathrm{i}}$, with $K^0_{eq,\mathrm{C4}} = \exp\left(-\left(\Delta_r G^0_{\mathrm{C4}} + 4F\Delta\Psi\right)/RT\right)$ and $\Delta_r G^0_{\mathrm{C4}} = \Delta_f G^0_{\mathrm{Cox}} + \Delta_f G^0_{\mathrm{H_2O}} - \Delta_r G^0_{\mathrm{Cred}} - \Delta_r G^0_{\mathrm{O_2}} = -202.2$ kJ/mol.

F_oF_1-ATPase flux:

$$J_{\mathrm{F1}} = X_{\mathrm{F1}}\left(K_{eq,\mathrm{F1}}[\mathrm{ADP}]_{\mathrm{x}}[\mathrm{PI}]_{\mathrm{x}} - [\mathrm{ATP}]_{\mathrm{x}}\right). \tag{B74}$$

where $K_{eq,\mathrm{F1}} = K^0_{eq,\mathrm{F1}} \cdot \frac{[\mathrm{H^+}]^{n_{\mathrm{A}}}_{\mathrm{i}}}{[\mathrm{H^+}]^{n_{\mathrm{A}}-1}_{\mathrm{x}}} \cdot \frac{P_{\mathrm{ATP}}}{P_{\mathrm{ADP}}P_{\mathrm{PI}}}$, with $K^0_{eq,\mathrm{F1}} = \exp\left(-\left(\Delta_r G^0_{\mathrm{F1}} - n_{\mathrm{A}}F\Delta\Psi\right)/RT\right)$ and $\Delta_r G^0_{\mathrm{F1}} = \Delta_f G^0_{\mathrm{ATP}} + \Delta_f G^0_{\mathrm{H_2O}} - \Delta_r G^0_{\mathrm{ADP}} - \Delta_r G^0_{\mathrm{PI}} = -4.51$ kJ/mol.

Mitochondrial adenylate kinase flux:

$$J_{\mathrm{AKi}} = X_{\mathrm{AK}}\left(K_{\mathrm{AK}}[\mathrm{ADP}]_{\mathrm{i}}[\mathrm{ADP}]_{\mathrm{i}} - [\mathrm{AMP}]_{\mathrm{i}}[\mathrm{ATP}]_{\mathrm{i}}\right). \tag{B75}$$

Mathematical expressions for TCA cycle fluxes

For brevity, values of Michaelis, inhibition, or activation constants are not presented here, but provided in Appendix C. In the following TCA cycle flux expressions, the enzyme activity $X_{\mathrm{TCA\ Enzyme}}$ in Table 3 in the paper is represented here by V_{mf}, and V_{mr} is related to V_{mf} by obeying Haledane equation (40).

Pyruvate dehydrogenase flux:

$$J_{\text{pdh}} = \frac{V_{mf}\left(1 - \frac{1}{K_{eq,\text{pdh}}}\frac{[\text{P}][\text{Q}][\text{R}]}{[\text{A}][\text{B}][\text{C}]}\right)}{K_{mC}\alpha_{i2}[\text{A}][\text{B}] + K_{mB}\alpha_{i1}[\text{A}][\text{C}] + K_{mA}[\text{B}][\text{C}] + [\text{A}][\text{B}][\text{C}]}, \tag{B76}$$

where [A] = [PYR], [B] = [COASH], [C] = [NAD], [P] = [CO_2tot], [Q] = [ACCOA], and [R] = [NADH].

Citrate Synthetase flux:

$$J_{\text{cits}} = \frac{V_{mf}\left([\text{A}][\text{B}] - \frac{[\text{P}][\text{Q}]}{K_{eq,\text{cits}}}\right)}{K_{ia}K_{mB}\alpha_{i1} + K_{mA}\alpha_{i1}[\text{B}] + K_{mB}\alpha_{i2}[\text{A}] + [\text{A}][\text{B}]}, \tag{B77}$$

where [A] = [OAA], [B] = [ACCOA], [P] = [COASH], and [Q] = [CIT].

Aconitase flux:

$$J_{\text{acon}} = \frac{V_{mf}V_{mr}\left([\text{A}] - \frac{[\text{P}]}{K_{eq,\text{acon}}}\right)}{K_{mA}V_{mr} + V_{mr}[\text{A}] + \frac{V_{mf}}{K_{eq,\text{acon}}}[\text{P}]}, \tag{B78}$$

where [A] = [CIT] and [P] = [ICIT].

Isocitrate dehydrogenase flux:

$$J_{\text{isod}} = \frac{V_{mf}\left(1 - \frac{1}{K_{eq,\text{isod}}}\frac{[\text{P}][\text{Q}][\text{R}]}{[\text{A}][\text{B}]}\right)}{1 + \left(\frac{K_{mB}}{[\text{B}]}\right)^{n_H}\alpha_i + \frac{K_{mA}}{[\text{A}]}\left(1 + \left(\frac{K_{ib}}{[\text{B}]}\right)^{n_H}\alpha_i + \frac{[\text{Q}]}{K_{iq}}\alpha_i\right)}, \tag{B79}$$

where [A] = [NAD], [B] = [ICIT], [P] = [AKG], [Q] = [NADH], and [R] = [CO_2tot].

α-Ketoglutarate dehydrogenase flux:

$$J_{\text{akgd}} = \frac{V_{mf}\left(1 - \frac{1}{K_{eq,\text{akgd}}}\frac{[\text{P}][\text{Q}][\text{R}]}{[\text{A}][\text{B}][\text{C}]}\right)}{\left(1 + \frac{K_{mA}}{[\text{A}]}\alpha_i + \frac{K_{mB}}{[\text{B}]}\left(1 + \frac{[\text{Q}]}{K_{iq}}\right) + \frac{K_{mC}}{[\text{C}]}\left(1 + \frac{[\text{R}]}{K_{ir}}\right)\right)}, \tag{B80}$$

where [A] = [AKG], [B] = [COASH], [C] = [NAD], [P] = [CO_2tot], [Q] = [SCOA], and [R] = [NADH].

Succinyl-CoA synthetase flux:

$$J_{\text{scoas}} = \frac{V_{mf}V_{mr}\left([\text{A}][\text{B}][\text{C}] - \frac{[\text{P}][\text{Q}][\text{R}]}{K_{eq,\text{scoas}}}\right)}{\begin{aligned}&V_{mr}K_{ia}K_{ib}K_{mC} + V_{mr}K_{ib}K_{mC}[\text{A}] + V_{mr}K_{ia}K_{mB}[\text{C}]\\ &+V_{mr}K_{mC}[\text{A}][\text{B}] + V_{mr}K_{mB}[\text{A}][\text{C}] + V_{mr}K_{mA}[\text{B}][\text{C}] + V_{mr}[\text{A}][\text{B}][\text{C}]\\ &+\frac{V_{mf}K_{ir}K_{mQ}[\text{P}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{iq}K_{mP}[\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mR}[\text{P}][\text{Q}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mQ}[\text{P}][\text{R}]}{K_{eq,\text{scoas}}}\\ &+\frac{V_{mf}K_{mP}[\text{Q}][\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}[\text{P}][\text{Q}][\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mQ}K_{ir}[\text{A}][\text{P}]}{K_{ia}K_{eq,\text{scoas}}} + \frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{R}]}{K_{ir}}\\ &+\frac{V_{mf}K_{mQ}K_{ir}[\text{A}][\text{B}][\text{Q}]}{K_{ia}K_{ib}K_{eq,\text{scoas}}} + \frac{V_{mr}K_{mA}[\text{B}][\text{C}][\text{R}]}{K_{ir}} + \frac{V_{mf}K_{mR}[\text{A}][\text{P}][\text{Q}]}{K_{ia}K_{eq,\text{scoas}}}\\ &+\frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{Q}][\text{R}]}{K_{iq}K_{ir}} + \frac{V_{mf}K_{ir}K_{mQ}[\text{A}][\text{B}][\text{C}][\text{P}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{ip}K_{mR}[\text{A}][\text{B}][\text{C}][\text{Q}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}}\\ &+\frac{V_{mf}K_{mR}[\text{A}][\text{B}][\text{P}][\text{Q}]}{K_{ia}K_{ib}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mA}[\text{B}][\text{C}][\text{Q}][\text{R}]}{K_{iq}K_{ir}} + \frac{V_{mr}K_{mA}K_{ic}[\text{B}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}}\\ &+\frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}} + \frac{V_{mf}K_{mR}[\text{A}][\text{B}][\text{C}][\text{P}][\text{Q}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mA}[\text{B}][\text{C}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}}\end{aligned}}, \quad \text{(B81)}$$

where [A] = [GDP], [B] = [SCOA], [C] = [PI], [P] = [COASH], [Q] = [SUC], and [R] = [GTP].

Succinate dehydrogenase flux:

$$J_{\text{sdh}} = \frac{V_{mf}V_{mr}\left([\text{A}][\text{B}] - \frac{[\text{P}][\text{Q}]}{K_{eq,\text{sdh}}}\right)}{\begin{aligned}&V_{mr}K_{ia}K_{mB}\alpha_i + V_{mr}K_{mB}[\text{A}] + V_{mr}K_{mA}\alpha_i[\text{B}] + \frac{V_{mf}K_{mQ}\alpha_i}{K_{eq,\text{sdh}}}[\text{P}] + \frac{V_{mf}K_{mP}}{K_{eq,\text{sdh}}}[\text{Q}]\\ &+V_{mr}[\text{A}][\text{B}] + \frac{V_{mf}K_{mQ}}{K_{eq,\text{sdh}}K_{ia}}[\text{A}][\text{P}] + \frac{V_{mr}K_{mA}}{K_{iq}}[\text{B}][\text{Q}] + \frac{V_{mf}}{K_{eq,\text{sdh}}}[\text{P}][\text{Q}]\end{aligned}}, \quad \text{(B82)}$$

where A] = [SUC], [B] = [COQ], [P] = [QH_2], and [Q] = [FUM].

Fumarase flux:

$$J_{\text{fum}} = \frac{V_{mf}V_{mr}\left([\text{A}] - \frac{[\text{P}]}{K_{eq,\text{fum}}}\right)}{K_{mA}V_{mr}\alpha_i + V_{mr}[\text{A}] + \frac{V_{mf}[\text{P}]}{K_{eq,\text{fum}}}}, \quad \text{(B83)}$$

where [A] = [FUM] and [P] = [MAL].

Malate dehydrogenase flux:

$$J_{\text{mdh}} = \frac{V_{mf} V_{mr} \left([\text{A}][\text{B}] - \dfrac{[\text{P}][\text{Q}]}{K_{eq,\text{mdh}}} \right)}{\begin{aligned} & V_{mr} K_{ia} K_{mB} \alpha_i + V_{mr} K_{mB} [\text{A}] + V_{mr} K_{mA} \alpha_i [\text{B}] + \frac{V_{mf} K_{mQ} \alpha_i [\text{P}]}{K_{eq,\text{mdh}}} + \frac{V_{mf} K_{mP} [\text{Q}]}{K_{eq,\text{mdh}}} \\ & + V_{mr} [\text{A}][\text{B}] + \frac{V_{mf} K_{mQ} [\text{A}][\text{P}]}{K_{eq,\text{mdh}} K_{ia}} + \frac{V_{mf} [\text{P}][\text{Q}]}{K_{eq,\text{mdh}}} + \frac{V_{mr} K_{mA} [\text{B}][\text{Q}]}{K_{iq}} \\ & + \frac{V_{mr} [\text{A}][\text{B}][\text{P}]}{K_{ip}} + \frac{V_{mf} [\text{B}][\text{P}][\text{Q}]}{K_{ib} K_{eq,\text{mdh}}} \end{aligned}}, \tag{B84}$$

where [A] = [NAD], [B] = [MAL], [P] = [OAA], and [Q] = [NADH].

Nucleoside diphosphokinase flux:

$$J_{\text{ndk}} = \frac{V_{mf} V_{mr} \left([\text{A}][\text{B}] - \dfrac{[\text{P}][\text{Q}]}{K_{eq,\text{ndk}}} \right) \Big/ \alpha_i}{\begin{aligned} & V_{mr} K_{mB} [\text{A}] + V_{mr} K_{mA} [\text{B}] + \frac{V_{mf} K_{mQ} [\text{P}]}{K_{eq,\text{ndk}}} + \frac{V_{mf} K_{mP} [\text{Q}]}{K_{eq,\text{ndk}}} + V_{mr} [\text{A}][\text{B}] \\ & + \frac{V_{mf} K_{mQ} [\text{A}][\text{P}]}{K_{eq,\text{ndk}} K_{ia}} + \frac{V_{mf} [\text{P}][\text{Q}]}{K_{eq,\text{ndk}}} + \frac{V_{mr} K_{mA} [\text{B}][\text{Q}]}{K_{iq}} \end{aligned}}, \tag{B85}$$

where [A] = [GTP], [B] = [ADP], [P] = [GDP], and [Q] = [ATP].

Glutamate oxaloacetate transaminase (aspartate transaminase) flux:

$$J_{\text{got}} = \frac{V_{mf} V_{mr} \left([\text{A}][\text{B}] - \dfrac{[\text{P}][\text{Q}]}{K_{eq,\text{got}}} \right)}{\begin{aligned} & V_{mr} K_{mB} [\text{A}] + V_{mr} K_{mA} \alpha_i [\text{B}] + \frac{V_{mf} K_{mQ} \alpha_i [\text{P}]}{K_{eq,\text{got}}} + \frac{V_{mf} K_{mP} [\text{Q}]}{K_{eq,\text{got}}} + V_{mr} [\text{A}][\text{B}] \\ & + \frac{V_{mf} K_{mQ} [\text{A}][\text{P}]}{K_{eq,\text{got}} K_{ia}} + \frac{V_{mf} [\text{P}][\text{Q}]}{K_{eq,\text{got}}} + \frac{V_{mr} K_{mA} [\text{B}][\text{Q}]}{K_{iq}} \end{aligned}}, \tag{B86}$$

where [A] = [ASP], [B] = [AKG], [P] = [OAA], and [Q] = [GLU].

Mathematical expressions for substrate and cation transport across the inner mitochondrial membrane

Adenine nucleotide translocase (ANT) flux:

$$J_{\text{ANT}} = \left(\frac{X_{\text{ANT}}}{1 + k_{m,\text{ADP}} / [\text{ADP}^{3-}]_{\text{i}}} \right) \left(\frac{[\text{ADP}^{3-}]_{\text{i}}}{[\text{ADP}^{3-}]_{\text{i}} + [\text{ATP}^{4-}]_{\text{i}} e^{(\theta - 1) F \Delta\Psi / RT}} - \frac{[\text{ADP}^{3-}]_{\text{x}}}{[\text{ADP}^{3-}]_{\text{x}} + [\text{ATP}^{4-}]_{\text{x}} e^{\theta F \Delta\Psi / RT}} \right), \tag{B87}$$

where $\theta = 0.6$.

Phosphate-hydrogen co-transporter flux:

$$J_{\mathrm{PIHt}} = X_{\mathrm{PIHt}} \cdot \frac{[\mathrm{H_2PO_4^-}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}} - [\mathrm{H_2PO_4^-}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}}}{k_{\mathrm{PIHt}}\left(1+[\mathrm{H_2PO_4^-}]_{\mathrm{i}}/k_{\mathrm{PIHt}}\right)\left(1+[\mathrm{H_2PO_4^-}]_{\mathrm{x}}/k_{\mathrm{PIHt}}\right)}. \tag{B88}$$

Potassium-hydrogen exchange flux:

$$J_{\mathrm{KH}} = X_{\mathrm{KH}}\left([\mathrm{K^+}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{x}} - [\mathrm{K^+}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{i}}\right). \tag{B89}$$

Pyruvate-hydrogen co-transporter flux:

$$J_{\mathrm{PYRH}} = X_{\mathrm{PYRH}}\left([\mathrm{PYR^-}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}} - [\mathrm{PYR^-}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}}\right). \tag{B90}$$

Glutamate-hydrogen co-transporter flux:

$$J_{\mathrm{GLUH}} = X_{\mathrm{GLUH}}\left([\mathrm{GLU^-}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}} - [\mathrm{GLU^-}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}}\right). \tag{B91}$$

Citrate-malate exchange flux:

$$J_{\mathrm{CITMAL}} = X_{\mathrm{CITMAL}}\left([\mathrm{HCIT^{2-}}]_{\mathrm{i}}[\mathrm{MAL^{2-}}]_{\mathrm{x}} - [\mathrm{HCIT^{2-}}]_{\mathrm{x}}[\mathrm{MAL^{2-}}]_{\mathrm{i}}\right). \tag{B92}$$

α-Ketoglutarate-malate exchange flux:

$$J_{\mathrm{AKGMAL}} = \frac{X_{\mathrm{AKGMAL}}\left([\mathrm{AKG}]_{\mathrm{i}}[\mathrm{MAL}]_{\mathrm{x}} - [\mathrm{AKG}]_{\mathrm{x}}[\mathrm{MAL}]_{\mathrm{i}}\right)}{K_{m\mathrm{AKGi}}K_{m\mathrm{MALx}}\left(2 + \frac{[\mathrm{MAL}]_{\mathrm{i}}}{K_{m\mathrm{MALi}}} + \frac{[\mathrm{MAL}]_{\mathrm{x}}}{K_{m\mathrm{MALx}}} + \frac{[\mathrm{AKG}]_{\mathrm{i}}}{K_{m\mathrm{AKGi}}} + \frac{[\mathrm{AKG}]_{\mathrm{x}}}{K_{m\mathrm{AKGx}}} + \frac{[\mathrm{MAL}]_{\mathrm{i}}[\mathrm{AKG}]_{\mathrm{x}}}{K_{m\mathrm{MALi}}K_{m\mathrm{AKGx}}} + \frac{[\mathrm{MAL}]_{\mathrm{x}}[\mathrm{AKG}]_{\mathrm{i}}}{K_{m\mathrm{MALx}}K_{m\mathrm{AKGi}}}\right)}, \tag{B93}$$

The above flux expression is derived based on rapid equilibrium random bi-bi mechanism proposed by Indiveri et al. (19), and the kinetic parameters are $K_{m\mathrm{MALi}}$ = 1.4 mM, $K_{m\mathrm{MALx}}$ = 0.7 mM, $\mathrm{K}_{\mathrm{mAKGi}}$ = 0.3 mM, and $K_{m\mathrm{AKGx}}$ = 0.17 mM.

Succinate/malate exchange flux:

$$J_{\mathrm{SUCMAL}} = X_{\mathrm{SUCMAL}}\left([\mathrm{SUC^{2-}}]_{\mathrm{i}}[\mathrm{MAL^{2-}}]_{\mathrm{x}} - [\mathrm{SUC^{2-}}]_{\mathrm{x}}[\mathrm{MAL^{2-}}]_{\mathrm{i}}\right). \tag{B94}$$

Malate/phosphate exchange flux:

$$J_{\mathrm{MALPI}} = X_{\mathrm{MALPI}}\left([\mathrm{MAL^{2-}}]_{\mathrm{i}}[\mathrm{PI^{2-}}]_{\mathrm{x}} - [\mathrm{MAL^{2-}}]_{\mathrm{x}}[\mathrm{PI^{2-}}]_{\mathrm{i}}\right). \tag{B95}$$

Aspartate-glutamate exchange flux:

$$J_{\mathrm{ASPGLU}} = \frac{X_{\mathrm{ASPGLU}}\left(K_{eq,\mathrm{ASPGLU}}[\mathrm{ASP}]_{\mathrm{i}}[\mathrm{GLU}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}} - [\mathrm{ASP}]_{\mathrm{x}}[\mathrm{GLU}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}}\right)}{\begin{aligned}&K_{eq,\mathrm{ASPGLU}}K_{i\mathrm{ASPi}}K_{i\mathrm{GLUx}}K_{h,\mathrm{ASPGLU}}\Bigg(2m + m\frac{[\mathrm{ASP}]_{\mathrm{i}}}{Ki\mathrm{ASPc}} + \frac{[\mathrm{ASP}]_{\mathrm{i}}[\mathrm{GLU}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}}}{K_{i\mathrm{ASPi}}K_{i\mathrm{GLUx}}K_{h,\mathrm{ASPGLU}}} \\ &+ m\frac{[\mathrm{ASP}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{i}}}{K_{i\mathrm{ASPx}}K_{h,\mathrm{ASPGLU}}} + \frac{[\mathrm{ASP}]_{\mathrm{x}}[\mathrm{GLU}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}}}{K_{i\mathrm{ASPx}}K_{i\mathrm{GLUi}}K_{h,\mathrm{ASPGLU}}} + m\frac{[\mathrm{ASP}]_{\mathrm{x}}}{K_{i\mathrm{ASPx}}} + m\frac{[\mathrm{ASP}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{x}}}{K_{i\mathrm{ASPi}}K_{h,\mathrm{ASPGLU}}} \\ &+ m\frac{[\mathrm{H^+}]_{\mathrm{x}}}{K_{h,\mathrm{ASPGLU}}} + m\frac{[\mathrm{GLU}]_{\mathrm{i}}[\mathrm{H^+}]_{\mathrm{i}}}{K_{i\mathrm{GLUi}}K_{h,\mathrm{ASPGLU}}} + m\frac{[\mathrm{H^+}]_{\mathrm{i}}}{K_{h,\mathrm{ASPGLU}}} + m\frac{[\mathrm{GLU}]_{\mathrm{x}}[\mathrm{H^+}]_{\mathrm{x}}}{K_{i\mathrm{GLUx}}K_{h,\mathrm{ASPGLU}}}\Bigg)\end{aligned}}, \tag{B96}$$

The above flux expression is derived based on rapid equilibrium random bi-bi mechanism with charge translocation proposed by Dierks et al. (12), and the kinetic parameters are $K_{i\mathrm{ASPi}}$ = 0.028 mM, $K_{i\mathrm{ASPx}}$ = 2.8 mM, $K_{i\mathrm{GLUi}}$ = 0.18 mM, $K_{i\mathrm{GLUx}}$ = 1.6 mM, $\mathrm{K}_{h,\mathrm{ASPGLU}} = 10^{-6.5}$, and m = 1.8.

Proton leak flux:

$$J_{\mathrm{Hle}} = X_{\mathrm{Hle}} \Delta\Psi \left(\frac{[\mathrm{H}^+]_\mathrm{c} e^{+F\Delta\Psi/RT} - [\mathrm{H}^+]_\mathrm{x}}{e^{+F\Delta\Psi/RT} - 1} \right). \tag{B97}$$

Mathematical expressions for passive permeation across the outer mitochondrial membrane

Adenine nucleoside permeation fluxes:

$$J_{\mathrm{ATPt}} = \gamma p_A \left([\mathrm{ATP}]_\mathrm{c} - [\mathrm{ATP}]_\mathrm{i} \right) \tag{B98}$$

$$J_{\mathrm{ADPt}} = \gamma p_A \left([\mathrm{ADP}]_\mathrm{c} - [\mathrm{ADP}]_\mathrm{i} \right) \tag{B99}$$

$$J_{\mathrm{AMPt}} = \gamma p_A \left([\mathrm{AMP}]_\mathrm{c} - [\mathrm{AMP}]_\mathrm{i} \right). \tag{B100}$$

Inorganic phosphate permeation flux:

$$J_{\mathrm{PIt}} = \gamma p_{\mathrm{PI}} \left([\mathrm{PI}]_\mathrm{c} - [\mathrm{PI}]_\mathrm{i} \right). \tag{B101}$$

TCA cycle intermediate permeation fluxes:

$$J_{\mathrm{PYRt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{PYR}]_\mathrm{c} - [\mathrm{PYR}]_\mathrm{i} \right) \tag{B102}$$

$$J_{\mathrm{CITt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{CIT}]_\mathrm{c} - [\mathrm{CIT}]_\mathrm{i} \right) \tag{B103}$$

$$J_{\mathrm{MALt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{MAL}]_\mathrm{c} - [\mathrm{MAL}]_\mathrm{i} \right) \tag{B104}$$

$$J_{\mathrm{AKGt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{AKG}]_\mathrm{c} - [\mathrm{AKG}]_\mathrm{i} \right) \tag{B105}$$

$$J_{\mathrm{SUCt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{SUC}]_\mathrm{c} - [\mathrm{SUC}]_\mathrm{i} \right) \tag{B106}$$

$$J_{\mathrm{GLUt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{GLU}]_\mathrm{c} - [\mathrm{GLU}]_\mathrm{i} \right) \tag{B107}$$

$$J_{\mathrm{ASPt}} = \gamma p_{\mathrm{TI}} \left([\mathrm{ASP}]_\mathrm{c} - [\mathrm{ASP}]_\mathrm{i} \right). \tag{B108}$$

Mathematical expressions for external space reaction fluxes

Hexokinase flux:

$$J_{\mathrm{HK}} = \frac{\mathrm{X}_{\mathrm{HK}} / \left(\mathrm{K}_{\mathrm{ib}} \mathrm{K}_{\mathrm{mA}} \right) \left([\mathrm{A}][\mathrm{B}] - [\mathrm{P}][\mathrm{Q}] / \mathrm{K}_{\mathrm{eq,HK}} \right)}{1 + \frac{[\mathrm{A}]}{\mathrm{K}_{\mathrm{ia}}} + \frac{[\mathrm{B}]}{\mathrm{K}_{\mathrm{ib}}} + \frac{[\mathrm{A}][\mathrm{B}]}{\mathrm{K}_{\mathrm{ib}} \mathrm{K}_{\mathrm{mA}}} + \frac{[\mathrm{P}]}{\mathrm{K}_{\mathrm{ip}}} + \frac{[\mathrm{Q}]}{\mathrm{K}_{\mathrm{iq}}} + \frac{[\mathrm{P}][\mathrm{Q}]}{\mathrm{K}_{\mathrm{iq}} \mathrm{K}_{\mathrm{mP}}} + \frac{[\mathrm{B}][\mathrm{P}]}{\mathrm{K}_{\mathrm{iG6P}} \mathrm{K}_{\mathrm{ib}}}}. \tag{B109}$$

where A denotes Mg^{2+}-bound ATP, B denotes GLC, P denotes G6P, and Q denotes Mg^{2+}-bound ADP.

The flux expression is adopted from Mulquiney and Kuchel's paper (33), and the parameters values are $\mathrm{K}_{\mathrm{mA}} = \mathrm{K}_{\mathrm{ia}}$ = 1 mM, $\mathrm{K}_{\mathrm{mB}} = \mathrm{K}_{\mathrm{ib}}$ = 1 mM, $\mathrm{K}_{\mathrm{mP}} = \mathrm{K}_{\mathrm{ip}}$ = 47 μM, $\mathrm{K}_{\mathrm{mQ}} = \mathrm{K}_{\mathrm{iq}}$ = 1 mM, and $\mathrm{K}_{\mathrm{iG6P}}$ = 1 μM.

Mitochondrial adenylate kinase flux:

$$J_{\mathrm{AKi}} = X_{\mathrm{AK}} \left(K_{\mathrm{AK}} [\mathrm{ADP}^{3-}]_\mathrm{i}^2 - [\mathrm{AMP}^{2-}]_\mathrm{i} [\mathrm{ATP}^{4-}]_\mathrm{i} \right), \tag{B110}$$

where X_{AK} is an large arbitrary value to maintain the reaction around equilibrium.

Cytoplasmic adenylate kinase flux:

$$J_{AKc} = X_{AK}\left(K_{AK}[ADP^{3-}]_c^{\,2} - [AMP^{2-}]_c[ATP^{4-}]_c\right), \tag{B111}$$

where X_{AK} is an large arbitrary value to maintain the reaction around equilibrium.

Creatine kinase flux:

$$J_{CKc} = X_{CK}\left(K_{CK}[ADP^{3-}]_c[PCr^{2-}]_c[H^{+}]_c - [ATP^{4-}]_c[Cr^{0}]_c\right), \tag{B112}$$

where X_{CK} is an large arbitrary value to maintain the reaction around equilibrium.

Appendix C. TCA Cycle Enzyme Kinetics—Enzyme Mechanisms and Assignment of Parameter Values

C1. Overview of Our Approach

Our approach to develop a kinetic model of the TCA cycle is to adapt and build on the detailed models of Kohn et al. (1, 13, 25-29). These models, developed in the late 1970's and early 1980's provide the most detailed kinetic scheme (in terms of accounting for all substrates and products in the reactions and a great deal of allosteric regulatory mechanism) for TCA cycle that are available in the literature. However the published papers do not provide sufficient information to reproduce the Kohn-Achs-Garfinkle model. In fact, even with a partial copy of the BiOSSIM computer code that we obtained from Michael Kohn, there is insufficient information to unambiguously construct a computational model. However, by mining information from the Kohn-Achs-Garfinkle papers, Kohn et al.'s original sources, Kohn et al.'s computer code, and additional literature, we have constructed reasonable kinetic models for each enzyme in the system, as described below.

As described in the main paper (Wu et al. "Computer Modeling of Mitochondrial TCA Cycle, Oxidative Phosphorylation, and Electrophysiology", 2007), the activity of each enzyme is estimated by comparison to data from mitochondria isolated from liver and heart. The assignment of other kinetic parameters is done based on the Kohn et al. publications, computer code, original references, and calculations from thermodynamic constraints. This process sometimes involves making somewhat arbitrary choices. Generally, where parameter values reported in the Kohn et al. publications and code are both in agreement with the rage of values reported in the original source, we have adopted the values published by Kohn et al. If the values from either the code or the Kohn et al. publications do not match the original sources, we choose a value from the original source(s).

Yet, in addition to reporting and justifying our specific choices, this document compiles alternative choices of parameter values.

C2. Notation and Abbreviations

The reactants and reference species for all metabolites in the system are listed in Table A1 in Appendix A. Charges for all of the reference species are indicated using superscripts, even when the charge is zero. This notation distinguishes biochemical reactants (for example, NADH) from references species (for example $NADH^0$).

Brackets are used to indicate concentration. For example [ATP] denotes the overall concentration of all species of the reactant ATP. A lowercase letter f is used to indicate total concentration of a reactant not bound to magnesium ion. For example

$$[ATP] = [ATP^{4-}] + [HATP^{3-}] + [KATP^{3-}] + [MgATP^{2-}]$$

and

$$[fATP] = [ATP^{4-}] + [HATP^{3-}] + [KATP^{3-}].$$

Binding polynomials are represented using P, where the subscript indicates the reactant. For example,

$$P_{\mathrm{ATP}} = 1 + \frac{[\mathrm{H}^+]}{K_{\mathrm{H\text{-}ATP}}} + \frac{[\mathrm{K}^+]}{K_{\mathrm{K\text{-}ATP}}} + \frac{[\mathrm{Mg}^+]}{K_{\mathrm{Mg\text{-}ATP}}}.$$

C3. Enzyme Mechanisms and Flux Expressions

1. Pyruvate Dehydrogenase

The reference reaction is

$$\mathrm{PYR^- + COAS^- + NAD^+ + H_2O \rightleftharpoons CO_3^{2-} + H^+ + ACCOA^0 + NADH^0},$$

and the biochemical reaction is

$$\mathrm{PYR + COASH + NAD + H_2O \rightleftharpoons CO_2tot + ACCOA + NADH}.$$

The standard Gibbs free energy is computed:

$$\Delta_r G^0_{\mathrm{pdh}} = \Delta_f G^0_{\mathrm{CO_2tot}} + \Delta_f G^0_{\mathrm{ACCOA}} + \Delta_f G^0_{\mathrm{NADH}} - \Delta_f G^0_{\mathrm{PYR}} - \Delta_f G^0_{\mathrm{COASH}} - \Delta_f G^0_{\mathrm{NAD}} - \Delta_f G^0_{\mathrm{H_2O}}.$$
$$= 19.59 \text{ kJ/mol}$$

The equilibrium constant for the reference reaction is

$$K^0_{eq,\mathrm{pdh}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{pdh}}}{RT}\right) = 5.02\times 10^{-4}\ \mathrm{M^{-1}}.$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{pdh}} = K^0_{eq,\mathrm{pdh}} \frac{1}{[\mathrm{H^+}]} \frac{P_{\mathrm{CO_2tot}} P_{\mathrm{ACCOA}} P_{\mathrm{NADH}}}{P_{\mathrm{PYR}} P_{\mathrm{COASH}} P_{\mathrm{NAD}}}.$$

The reaction mechanism is assumed to be hexa-uni-ping-pong when the reactant water is ignored (9), with the following fundamental steps (27):

$$\mathrm{E + PYR^-} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E\cdot PYR^-}$$

$$\mathrm{E\cdot PYR^-} \xrightarrow{k_2} \mathrm{E\cdot CHOCH_3^- + CO_2}$$

$$\mathrm{E\cdot CHOCH_3^- + COASH^0} \underset{\mathrm{k_{-3}}}{\overset{\mathrm{k_{+3}}}{\rightleftharpoons}} \mathrm{E\cdot ACCOA^-}$$

$$\mathrm{E\cdot ACCOA^-} \xrightarrow{k_4} \mathrm{E^- + ACCOA^0}$$

$$\mathrm{E^- + NAD^+} \underset{k_{-5}}{\overset{k_{+5}}{\rightleftharpoons}} \mathrm{E\cdot NAD^0}$$

$$\mathrm{E\cdot NAD^0} \xrightarrow{k_6} \mathrm{E + NADH^0}$$

The net reaction for this mechanism, in terms of chemical species is

$$\mathrm{PYR^- + COASH^0 + NAD^+ \rightleftharpoons CO_2 + ACCOA^0 + NADH^0}.$$

The flux expression for this mechanism follows from application of the King-Altman method (23, 41). In addition to the reaction steps shown above, in Kohn et al's model, ACCOA and NADH competitively inhibit enzymatic activity against COASH and NAD with binding constants $K_{i\mathrm{ACCOA}}$ and $K_{i\mathrm{COASH}}$, respectively (27).

Denoting the reactant concentrations as [A] = [PYR], [B] = [COASH], [C] = [NAD], [P] = [CO_2tot], [Q] = [ACCOA], and [R] = [NADH], we have

$$J^+_{\mathrm{pdh}} = \frac{V_{mf}[\mathrm{A}][\mathrm{B}][\mathrm{C}]}{K_{mC}\alpha_{i2}[\mathrm{A}][\mathrm{B}] + K_{mB}\alpha_{i1}[\mathrm{A}][\mathrm{C}] + K_{mA}[\mathrm{B}][\mathrm{C}] + [\mathrm{A}][\mathrm{B}][\mathrm{C}]},$$

where

$$K_{mA} = \frac{k_4 k_6 (k_{-1} + k_2)}{k_{+1}(k_2 k_4 + k_2 k_6 + k_4 k_6)}, \; K_{mB} = \frac{k_2 k_6 (k_{-3} + k_4)}{k_{+3}(k_2 k_4 + k_2 k_6 + k_4 k_6)}, \; K_{mC} = \frac{k_2 k_4 (k_{-5} + k_6)}{k_{+5}(k_2 k_4 + k_2 k_6 + k_4 k_6)},$$

$$V_{mf} = [\mathrm{E}]_t \frac{k_2 k_4 k_6}{k_2 k_4 + k_2 k_6 + k_4 k_6},$$

$$\alpha_{i1} = 1 + \frac{[\mathrm{ACCOA}]}{K_{i\mathrm{ACCOA}}}, \alpha_{i2} = 1 + \frac{[\mathrm{NADH}]}{K_{i\mathrm{NADH}}}.$$

The above expression for J^{+}_{pdh} is the forward flux of the pyruvate dehydrogenase reaction. To account for the Haldane constraints (41), the flux expression is modified as follows:

$$J_{\mathrm{pdh}} = J^{+}_{\mathrm{pdh}} - J^{-}_{\mathrm{pdh}} = J^{+}_{\mathrm{pdh}} \left(1 - \frac{1}{K_{eq,\mathrm{pdh}}} \frac{[\mathrm{P}][\mathrm{Q}][\mathrm{R}]}{[\mathrm{A}][\mathrm{B}][\mathrm{C}]} \right).$$

In Kohn et al's paper, the V_{mf} is assumed to depend on the current and historic activities of pyruvate dehydrogenase kinase (PDHK) and pyruvate dehydrogenase phosphate phosphatase (PDHP) (27). However, in the current model, the pyruvate dehydrogenase complex (PDH) activity, i.e. V_{mf}, is assumed to be a constant to simplify PDH flux modeling.

Parameter Values for Pyruvate Dehydrogenase

Parameter	Kohn et al. (27)[a] (μM)	Tsai et al. (48)[b] (μM)	Model values (μM)
K_{mA}	38.3	42	38.3
K_{mB}	9.9	20	9.9
K_{mC}	60.7	78	60.7
$K_{i\mathrm{ACCOA}}$	40.2	--	40.2
$K_{i\mathrm{NADH}}$	40.0	--	40.0

[a] Averages of experimental values reported for rat heart and kidney, pig heart, and bovine heart in the literature.
[b] Values from bovine kidney.

2. Citrate Synthase

The reference reaction is

$$\mathrm{OAA}^{2-} + \mathrm{ACCOA}^{0} + \mathrm{H_2O} \rightleftharpoons \mathrm{COAS}^{-} + \mathrm{CIT}^{3-} + 2\mathrm{H}^{+},$$

and the biochemical reaction is

$$\mathrm{OAA} + \mathrm{ACCOA} + \mathrm{H_2O} \rightleftharpoons \mathrm{COASH} + \mathrm{CIT}.$$

The standard Gibbs free energy is computed:

$$\Delta_r G^0_{\mathrm{cits}} = \Delta_f G^0_{\mathrm{COASH}} + \Delta_f G^0_{\mathrm{CIT}} - \Delta_f G^0_{\mathrm{OAA}} - \Delta_f G^0_{\mathrm{ACCOA}} - \Delta_f G^0_{\mathrm{H_2O}} = 42.36 \text{ kJ/mol}.$$

The equilibrium constant for the reference reaction is

$$K^0_{eq,\text{cits}} = \exp\left(-\frac{\Delta_r G^0_{\text{cits}}}{RT}\right) = 7.34\times 10^{-8}\ \text{M}^{-1}.$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\text{cits}} = K^0_{eq,\text{cits}}\frac{1}{[\text{H}^+]^2}\frac{P_{\text{COA}}P_{\text{CIT}}}{P_{\text{OAA}}P_{\text{ACCOA}}}.$$

The reaction mechanism is assumed to be ordered bi-bi when the reactant water is ignored (9), with the following fundamental steps (26):

$$\text{E} + \text{OAA}^{2-} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \text{E}\cdot\text{OAA}^{2-}$$

$$\text{E}\cdot\text{OAA}^{2-} + \text{ACCOA}^0 \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \text{E}\cdot\text{OAA}^{2-}\cdot\text{ACCOA}^0$$

$$\text{E}\cdot\text{OAA}^{2-}\cdot\text{ACCOA}^0 + \text{H}_2\text{O} \xrightarrow{k_3} \text{E}\cdot\text{HCIT}^{2-} + \text{COASH}^0$$

$$\text{E}\cdot\text{HCIT}^{2-} \xrightarrow{k_4} \text{E} + \text{CIT}^{3-} + \text{H}^+.$$

The net reaction for this mechanism, in terms of chemical species is

$$\text{OAA}^{2-} + \text{ACCOA}^0 + \text{H}_2\text{O} \rightleftharpoons \text{COASH}^0 + \text{CIT}^{3-}.$$

In addition to the reaction steps shown above, unchelated citrate (uCIT) acts as a competitive inhibitor against OAA^{2-} with a binding constant $K_{i\text{CIT}}$, and unchelated adenine nucleotides (uATP, uADP, and uAMP), COASH, and succinyl COA are uncompetitive inhibitors against ACCOA^0 with binding constants $K_{i\text{ATP}}$, $K_{i\text{ADP}}$, $K_{i\text{AMP}}$, $K_{i\text{COA}}$, and $K_{i\text{SCOA}}$ respectively (26).

The overall flux expression is derived from steady state mass balance. Denoting the reactant concentrations as [A] = [OAA], [B] = [ACCOA], [P] = [COASH], and [Q] = [CIT], we have

$$J^+_{cits} = \frac{V_{mf}[\text{A}][\text{B}]}{K_{ia}K_{mB}\alpha_{i1} + K_{mA}\alpha_{i1}[\text{B}] + K_{mB}\alpha_{i2}[\text{A}] + [\text{A}][\text{B}]},$$

where

$$K_{mA} = \frac{k_3k_4}{k_{+1}(k_3+k_4)},\ K_{mB} = \frac{k_4(k_{-2}+k_3)}{k_{+2}(k_3+k_4)},\ K_{ia} = \frac{k_{-1}}{k_{+1}},\ V_{mf} = [\text{E}]_t\frac{k_3k_4}{(k_3+k_4)},$$

$$\alpha_{i1} = 1 + \frac{[\text{uCIT}]}{K_{i\text{CIT}}}, \alpha_{i2} = 1 + \frac{[\text{uATP}]}{K_{i\text{ATP}}} + \frac{[\text{uADP}]}{K_{i\text{ADP}}} + \frac{[\text{uAMP}]}{K_{i\text{AMP}}} + \frac{[\text{COASH}]}{K_{i\text{COASH}}} + \frac{[\text{SCOA}]}{K_{i\text{SCOA}}}.$$

The above expression for is the forward flux of the citrate synthetase reaction; the functional form is consistent with ordered bi-bi rate equation if inhibition coefficients α_{i1} and α_{i2} are set to 1 (41). To account for the Haldane constraints (41), the flux expression is modified as follows:

$$J_{\text{cits}} = J^+_{\text{cits}} - J^-_{\text{cits}} = J^+_{\text{cits}}\left(1 - \frac{1}{K_{eq,\text{cits}}}\frac{[\text{P}][\text{Q}]}{[\text{A}][\text{B}]}\right).$$

Parameter Values for Citrate Synthase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Shepherd and Garland (42) (μM)	Smith and Williamson (45) (μM)	Model values (μM)
K_{mA}	4	4	2[a]	1.6[b]	4
K_{mB}	7	14	16[a]	8[b]-10[a]	14
K_{ia}	3.33	--	--	--	3.33
$K_{i\mathrm{CIT}}$	--	1600	--	1600[b]	1600
$K_{i\mathrm{ATP}}$	--	900	550[a]	950[b]	900
$K_{i\mathrm{ADP}}$	--	1800	1400[a]	--	1800
$K_{i\mathrm{AMP}}$	--	6000	6700[a]	--	6000
$K_{i\mathrm{COASH}}$	--	675	--	67[b]	67[c]
$K_{i\mathrm{SCOA}}$	--	140	--	130[b]-140[a]	140[d]

[a] Values obtained from rat liver.
[b] Values from bovine heart.
[c] Obvious difference exists between Kohn et al.'s value and the original source. It is pointed out that "… its (CoA-SH inhibition) strength relative to that of the inhibition by acyl-CoA derivatives may vary greatly with the conditions used for kinetic measurements. …" (45). Since Kohn et al. did not provide an explicit source of their Ki_{COASH} value, we choose to use the experimental data from Smith and Williamson.
[d] This value is presented as 1.4 mM in Kohn et al.'s earlier paper (26), but it is pointed out to be a typo and corrected to be 140 mM in their later publication (29).

3. Aconitase

The reference reaction is

$$\mathrm{CIT}^{3-} \rightleftharpoons \mathrm{ICIT}^{3-},$$

and the biochemical reaction is

$$\mathrm{CIT} \rightleftharpoons \mathrm{ICIT}.$$

The equilibrium constant for the reference reaction is:

$$K^0_{eq,\mathrm{acon}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{acon}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\mathrm{ICIT}} - \Delta_f G^0_{\mathrm{CIT}}}{RT}\right) = 7.59\times10^{-2} \quad .$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{acon}} = K^0_{eq,\mathrm{acon}} \frac{P_{\mathrm{ICIT}}}{P_{\mathrm{CIT}}}.$$

Kohn et al. (26) assume that both magnesium-bound and unbound species are involved in the reaction mechanism. However, the kinetic constants are derived from a single set of experiments in which divalent cation was provided in excess (46). Therefore the model can be equivalently formulated in terms of an ordered uni-uni mechanism for total citrate and isocitrate.

The fundamental reaction steps are

$$E + CIT^{3-} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} E \cdot CIT^{3-}$$

$$E \cdot CIT^{3-} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} E + ICIT^{3-}$$

The net reaction for this mechanism, in terms of chemical species is

$$CIT^{3-} \rightleftharpoons ICIT^{3-}.$$

The flux expression for this mechanism is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [CIT] and [P] = [ICIT], we have

$$J_{\text{acon}} = \frac{\frac{V_{mf}}{K_{mA}}[A] - \frac{V_{mr}}{K_{mP}}[P]}{1 + \frac{[A]}{K_{mA}} + \frac{[P]}{K_{mP}}} = \frac{V_{mf}V_{mr}\left([A] - \frac{[P]}{K_{eq,\text{acon}}}\right)}{K_{mA}V_{mr} + V_{mr}[A] + \frac{V_{mf}}{K_{eq,\text{acon}}}[P]}.$$

where

$$K_{mA} = \frac{k_{-1} + k_{+2}}{k_{+1}},\ K_{mP} = \frac{k_{-1} + k_{+2}}{k_{-2}},\ V_{mf} = [E]_t k_{+2},\ V_{mr} = [E]_t k_{-1}.$$

Parameter Values for Aconitase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Thomson et al. (46) (μM)	Model values (μM)
K_{mA}	1100	1100[a]	950[b]	1161[c]
K_{mP}	295.6	480[a]	140[b]	434[c]

[a] These values are computed for unchelated citrate and iso-citrate from Thomson et al.'s K_m values (assumed to be the K_m values of the metal chelates) (46) by using Blair's magnesium binding constants (6), respectively (26).
[b] These values are measured in the presence of excess divalent metal (46).
[c] These values have been corrected to apply to total reactant concentration for citrate and iso-citrate by using Blair's magnesium binding constants (6), at a physiologically reasonably free magnesium concentration of $[Mg^{2+}]$ = 1 mM.

4. Isocitrate Dehydrogenase

The reference reaction is

$$NAD^{+} + ICIT^{3-} + H_2O \rightleftharpoons AKG^{2-} + NADH^{0} + CO_3^{2-} + 2H^{+},$$

and the biochemical reaction is

$$\text{NAD} + \text{ICIT} + \text{H}_2\text{O} \rightleftharpoons \text{AKG} + \text{NADH} + \text{CO}_2\text{tot}\,.$$

The equilibrium constant for the reference reaction is:

$$K^0_{eq,\text{isod}} = \exp\left(-\frac{\Delta_r G^0_{\text{isod}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\text{AKG}} + \Delta_f G^0_{\text{NADH}} + \Delta_f G^0_{\text{CO}_2\text{tot}} - \Delta_f G^0_{\text{NAD}} - \Delta_f G^0_{\text{ICIT}} - \Delta_f G^0_{\text{H}_2\text{O}}}{RT}\right)$$

$$= 3.50\times 10^{-16}\,.$$

The apparent equilibrium constants for the biochemical reaction are

$$K_{eq,\text{isod}} = K^0_{eq,\text{isod}}\,\frac{1}{[\text{H}^+]^2}\,\frac{P_{\text{AKG}}P_{\text{NADH}}P_{\text{CO}_2\text{tot}}}{P_{\text{NAD}}P_{\text{ICIT}}}\,.$$

The reaction mechanism is assumed to be ordered bi-ter with the following fundamental steps (26)

$$\text{E} + \text{NAD}^+ \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \text{E}\cdot\text{NAD}^+$$

$$\text{E}\cdot\text{NAD}^+ + \text{MgICIT}^- \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \text{E}\cdot\text{NAD}\cdot\text{MgICIT}^0$$

$$\text{E}\cdot\text{NAD}\cdot\text{MgCIT} \underset{k_{-4}}{\overset{k_{+4}}{\rightleftharpoons}} \text{E} + \text{AKG}^{2-} + \text{CO}_2 + \text{NADH}^0 + \text{Mg}^{2+}\,.$$

The net reaction for this mechanism, in terms of chemical species is

$$\text{NAD}^+ + \text{MgICIT}^- \rightleftharpoons \text{AKG}^{2-} + \text{CO}_2 + \text{NADH}^0 + \text{Mg}^{2+}\,.$$

Kohn et al. realized that the above mechanism "was a greatly simplified empirical representation of the complex kinetics of this allosteric enzyme", and they provided the flux expression with inhibition and activation in their later publication (29).

We have corrected the apparent K_i for ATP and K_a for ADP to correspond to reactant (sum of species) concentrations. At a physiologically reasonably free magnesium concentration of $[\text{Mg}^{2+}]$ = 1 mM, approximately 90% and 50% of total ATP and ADP are in the magnesium bound-forms, respectively. Therefore, to scale K_i and K_a that apply to magnesium-bound species to apply to total reactant concentrations, the K_i for ATP and K_a for ADP are multiplied by 1/0.1 and 1/0.5, respectively.

Denoting the reactant concentrations as [A] = [NAD], [B] = [ICIT], [P] = [AKG], [Q] = [NADH], and [R] = [CO_2tot], we have

$$J^+_{\text{isod}} = \frac{V_{mf}}{1 + \left(\frac{K_{mB}}{[\text{B}]}\right)^{n_H}\alpha_i + \frac{K_{mA}}{[\text{A}]}\left[1 + \left(\frac{K_{ib}}{[\text{B}]}\right)^{n_H}\alpha_i + \frac{[\text{Q}]}{K_{iq}}\alpha_i\right]}$$

$$= \frac{V_{mf}[\text{A}][\text{B}]^{n_H}}{[\text{A}][\text{B}]^{n_H} + K_{mB}{}^{n_H}\alpha_i[\text{A}] + K_{mA}\left[[\text{B}]^{n_H} + K_{ib}{}^{n_H}\alpha_i + \frac{[\text{Q}][\text{B}]^{n_H}}{K_{iq}}\alpha_i\right]}$$

where

$$\alpha_i = 1 + \frac{K_{a\mathrm{ADP}}}{[\mathrm{fADP}]}\left(1 + \frac{[\mathrm{fATP}]}{K_{i\mathrm{ATP}}}\right).$$

To account for the Haldane constraints (41), the flux expression is modified as follows:

$$J_{\mathrm{isod}} = J^{+}_{\mathrm{isod}} - J^{-}_{\mathrm{isod}} = J^{+}_{\mathrm{isod}}\left(1 - \frac{1}{K_{eq,\mathrm{isod}}}\frac{[\mathrm{P}][\mathrm{Q}][\mathrm{R}]}{[\mathrm{A}][\mathrm{B}]}\right)$$

$$= \frac{V_{mf}\left([\mathrm{A}][\mathrm{B}]^{n_H} - \dfrac{[\mathrm{B}]^{n_H - 1}[\mathrm{P}][\mathrm{Q}][\mathrm{R}]}{K_{eq,\mathrm{isod}}}\right)}{[\mathrm{A}][\mathrm{B}]^{n_H} + K_{mB}{}^{n_H}\alpha_i[\mathrm{A}] + K_{mA}\left[[\mathrm{B}]^{n_H} + K_{ib}{}^{n_H}\alpha_i + \dfrac{[\mathrm{Q}][\mathrm{B}]^{n_H}}{K_{iq}}\alpha_i\right]}.$$

Parameter Values for Isocitrate Dehydrogenase

Parameter	Kohn et al. (26, 29) (μM)	Denton et al. (10) (μM)	Plaut et al. (39) (μM)	Model values (μM)
K_{mA}	74	--	70[b]	74
K_{mB}	59	53[a]	--	183[c]
n_H	3.0	3.3[a]	--	3.0 (unitless)
K_{ib}	7.66	--	--	23.8[c]
K_{iq}	29	--	--	29
$K_{i\mathrm{ATP}}$	91	--	118[b]	91
$K_{a\mathrm{ADP}}$	50	--	67-544[b]	50

[a] These values are measured from rat heart mitochondria.
[b] These values are measured from bovine heart.
[c] These values have been corrected to apply to total reactant concentration for isocitrate (refer to the Aconitase part), at a physiologically reasonably free magnesium concentration of $[\mathrm{Mg}^{2+}]$ = 1 mM.

5. α-Ketoglutarate dehydrogenase

The reference reaction is

$$\mathrm{AKG}^{2-} + \mathrm{COAS}^{-} + \mathrm{NAD}^{+} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO}_3^{2-} + \mathrm{SCOA}^{-} + \mathrm{NADH}^{0} + \mathrm{H}^{+},$$

and the biochemical reaction is

$$\mathrm{AKG} + \mathrm{COASH} + \mathrm{NAD} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2tot} + \mathrm{SCOA} + \mathrm{NADH} \ .$$

The equilibrium constant for the reference reaction is

$$K^0_{eq,\text{akgd}} = \exp\left(-\frac{\Delta_r G^0_{\text{akgd}}}{RT}\right)$$

$$= \exp\left(-\frac{\Delta_f G^0_{\text{CO2tot}} + \Delta_f G^0_{\text{SCOA}} + \Delta_f G^0_{\text{NADH}} - \Delta_f G^0_{\text{AKG}} - \Delta_f G^0_{\text{COASH}} - \Delta_f G^0_{\text{NAD}} - \Delta_f G^0_{\text{H}_2\text{O}}}{RT}\right)$$

$$= 6.93\times 10^{-3} \ .$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\text{akgd}} = K^0_{eq,\text{akgd}} \frac{1}{[\text{H}^+]} \frac{P_{\text{CO}_2\text{tot}} P_{\text{SCOA}} P_{\text{NADH}}}{P_{\text{AKG}} P_{\text{COASH}} P_{\text{NAD}}} \ .$$

The reaction mechanism is assumed to be hexa-uni-ping-pong-ter-ter (41) with the following fundamental steps (26):

$$\text{E} + \text{AKG}^{2-} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \text{E}\cdot\text{AKG}^{2-}$$

$$\text{E}\cdot\text{AKG}^{2-} \xrightarrow{k_2} \text{E}\cdot\text{SUC}^{2-} + \text{CO}_2$$

$$\text{E}\cdot\text{SUC}^{2-} + \text{COASH}^0 \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} \text{E}\cdot\text{SCOA}^{2-}$$

$$\text{E}\cdot\text{SCOA}^{2-} \xrightarrow{k_4} \text{E}^- + \text{SCOA}^-$$

$$\text{E}^- + \text{NAD}^+ \underset{k_{-5}}{\overset{k_{+5}}{\rightleftharpoons}} \text{E}\cdot\text{NADH}$$

$$\text{E}\cdot\text{NADH} \xrightarrow{k_6} \text{E} + \text{NADH}^0 .$$

The net reaction for this mechanism, in terms of chemical species is

$$\text{AKG}^{2-} + \text{COASH}^0 + \text{NAD}^+ \rightleftharpoons \text{CO}_2 + \text{SCOA}^- + \text{NADH}^0 .$$

In Kohn and Garfinkel's later publication (29), the flux expression does not strictly follow the above reaction mechanism, and the expression includes inhibition and activation. Denoting the reactant concentrations as [A] = [AKG], [B] = [COASH], [C] = [NAD], [P] = [CO_2tot], [Q] = [SCOA], and [R] = [NADH], we have

$$J^+_{\text{akgd}} = \frac{V_{mf}}{\left[1 + \frac{K_{mA}}{[\text{A}]}\alpha_i + \frac{K_{mB}}{[\text{B}]}\left(1 + \frac{[\text{Q}]}{K_{iq}}\right) + \frac{K_{mC}}{[\text{C}]}\left(1 + \frac{[\text{R}]}{K_{ir}}\right)\right]}$$

where

$$\alpha_i = 1 + \frac{K_{a\text{ADP}}}{[\text{fADP}]}\left(1 + \frac{[\text{fATP}]}{K_{i\text{ATP}}}\right).$$

To account for the Haldane constraints (41), the flux expression is modified as follows:

$$J_{\text{akgd}} = J^+_{\text{akgd}}\left(1 - \frac{1}{K_{eq,\text{akgd}}} \frac{[\text{P}][\text{Q}][\text{R}]}{[\text{A}][\text{B}][\text{C}]}\right).$$

Parameter Values for α-Ketoglutarate Dehydrogenase

Parameter	Kohn et al. (26, 29) (μM)	Williamson et al. (54) (μM)	Smith et al. (44) (μM)	Model values (μM)
K_{mA}	80	80[c]	--	80
K_{mB}	55[a]	--	2.7[d]	55
K_{mC}	21	21[d]	21[d]	21
K_{iq}	6.9	--	3.9-6.9[d]	6.9
K_{ir}	4.5	4.5[d]	4.5[d]	0.60[e]
$K_{i\mathrm{ATP}}$	50[b]	--	--	50
$K_{a\mathrm{ADP}}$	100[b]	--	--	100

[a] This value has been increased to include levels of fatty acyl CoA thioesters into CoASH and succinyl CoA pools.
[b] These value have been obtained based on the data of McCormack and Denton (32). $K_{i\mathrm{ATP}}$ has been calculated from an expression $K_{a\mathrm{ADP}}^{app} = K_{a\mathrm{ADP}}\left(1+\frac{[\mathrm{ATP}]}{K_{i\mathrm{ATP}}}\right)$ at very low divalent cation concentration ($[\mathrm{Ca}^{2+}]$ < 1nM and $[\mathrm{Mg}^{2+}] = \sim 0$), and thus it applies to unchelated ATP.
[c] These values are measured from isolated rat heart mitochondria.
[d] These values are measured from pig heart.
[e] This value are obtained based on optimization to match model predictions with LaNoue et al.'s experimental measurements (30).

6. Succinyl-CoA Synthetase

The reference reaction is

$$\mathrm{GDP}^{3-} + \mathrm{SCOA}^{-} + \mathrm{PI}^{2-} \rightleftharpoons \mathrm{COAS}^{-} + \mathrm{SUC}^{2-} + \mathrm{GTP}^{4-} + \mathrm{H}^{+},$$

and the biochemical reaction is

$$\mathrm{GDP} + \mathrm{SCOA} + \mathrm{PI} \rightleftharpoons \mathrm{COASH} + \mathrm{SUC} + \mathrm{GTP}.$$

The equilibrium constant of the reference reaction is:

$$K_{eq,\mathrm{scoas}}^{0} = \exp\left(-\frac{\Delta_r G_{\mathrm{scoas}}^{0}}{RT}\right)$$
$$= \exp\left(-\frac{\Delta_f G_{\mathrm{COASH}}^{0} + \Delta_f G_{\mathrm{SUC}}^{0} + \Delta_f G_{\mathrm{GTP}}^{0} - \Delta_f G_{\mathrm{GDP}}^{0} - \Delta_f G_{\mathrm{SCOA}}^{0} - \Delta_f G_{\mathrm{PI}}^{0}}{RT}\right) = 9.54\times10^{-9}\ \mathrm{M}^{-1}.$$

The apparent equilibrium constant for the biochemical reaction is:

$$K_{eq,\mathrm{scoas}} = K_{eq,\mathrm{scoas}}^{0}\frac{1}{[\mathrm{H}^{+}]}\frac{P_{\mathrm{COASH}}P_{\mathrm{SUC}}P_{\mathrm{GTP}}}{P_{\mathrm{GDP}}P_{\mathrm{SCOA}}P_{\mathrm{PI}}}.$$

The reaction mechanism is assumed to be ordered ter-ter, with the following fundamental steps (26):

$$E + MgGDP^- \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} E \cdot MgGDP^-$$

$$E \cdot MgGDP^- + SCOA^- \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} E \cdot MgGDP^- \cdot SCOA^-$$

$$E \cdot MgGDP^- \cdot SCOA^- + PI^{2-} \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} E \cdot MgGDP^- \cdot SCOA^- \cdot PI^{2-}$$

$$E \cdot MgGDP^- \cdot SCOA^- \cdot PI^{2-} \underset{k_{-4}}{\overset{k_{+4}}{\rightleftharpoons}} E \cdot MgGTP^{2-} \cdot SUC^{2-} + COASH$$

$$E \cdot MgGTP^{2-} \cdot SUC^{2-} \underset{k_{-5}}{\overset{k_{+5}}{\rightleftharpoons}} E \cdot MgGTP^{2-} + SUC^{2-}$$

$$E \cdot MgGTP^{2-} \underset{k_{-6}}{\overset{k_{+6}}{\rightleftharpoons}} E + MgGTP^{2-}$$

The net reaction for this mechanism, in terms of chemical species is

$$MgGDP^- + SCOA^- + PI^{2-} \rightleftharpoons COASH + SUC^{2-} + MgGTP^{2-} .$$

The flux expression for this mechanism follows from application of the King-Altman method (23, 41). Denoting the reactant concentrations as [A] = [GDP], [B] = [SCOA], [C] = [PI], [P] = [COASH], [Q] = [SUC], and [R] = [GTP], we have

$$J_{\text{scoas}} = \frac{V_{mf}V_{mr}\left([\text{A}][\text{B}][\text{C}] - \dfrac{[\text{P}][\text{Q}][\text{R}]}{K_{eq,\text{scoas}}}\right)}{\begin{aligned} & V_{mr}K_{ia}K_{ib}K_{mC} + V_{mr}K_{ib}K_{mC}[\text{A}] + V_{mr}K_{ia}K_{mB}[\text{C}] \\ & + V_{mr}K_{mC}[\text{A}][\text{B}] + V_{mr}K_{mB}[\text{A}][\text{C}] + V_{mr}K_{mA}[\text{B}][\text{C}] + V_{mr}[\text{A}][\text{B}][\text{C}] \\ & + \frac{V_{mf}K_{ir}K_{mQ}[\text{P}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{iq}K_{mP}[\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mR}[\text{P}][\text{Q}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mQ}[\text{P}][\text{R}]}{K_{eq,\text{scoas}}} \\ & + \frac{V_{mf}K_{mP}[\text{Q}][\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}[\text{P}][\text{Q}][\text{R}]}{K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mQ}K_{ir}[\text{A}][\text{P}]}{K_{ia}K_{eq,\text{scoas}}} + \frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{R}]}{K_{ir}} \\ & + \frac{V_{mf}K_{mQ}K_{ir}[\text{A}][\text{B}][\text{Q}]}{K_{ia}K_{ib}K_{eq,\text{scoas}}} + \frac{V_{mr}K_{mA}[\text{B}][\text{C}][\text{R}]}{K_{ir}} + \frac{V_{mf}K_{mR}[\text{A}][\text{P}][\text{Q}]}{K_{ia}K_{eq,\text{scoas}}} \\ & + \frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{Q}][\text{R}]}{K_{iq}K_{ir}} + \frac{V_{mf}K_{ir}K_{mQ}[\text{A}][\text{B}][\text{C}][\text{P}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{ip}K_{mR}[\text{A}][\text{B}][\text{C}][\text{Q}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}} \\ & + \frac{V_{mf}K_{mR}[\text{A}][\text{B}][\text{P}][\text{Q}]}{K_{ia}K_{ib}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mA}[\text{B}][\text{C}][\text{Q}][\text{R}]}{K_{iq}K_{ir}} + \frac{V_{mr}K_{mA}K_{ic}[\text{B}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}} \\ & + \frac{V_{mr}K_{ia}K_{mB}[\text{C}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}} + \frac{V_{mf}K_{mR}[\text{A}][\text{B}][\text{C}][\text{P}][\text{Q}]}{K_{ia}K_{ib}K_{ic}K_{eq,\text{scoas}}} + \frac{V_{mf}K_{mA}[\text{B}][\text{C}][\text{P}][\text{Q}][\text{R}]}{K_{ip}K_{iq}K_{ir}} \end{aligned}} ,$$

where

$$K_{mA} = \frac{k_{+4}k_{+5}k_{+6}}{k_{+1}(k_{+4}k_{+5} + k_{+4}k_{+6} + k_{+5}k_{+6})}, \; K_{mB} = \frac{k_{+4}k_{+5}k_{+6}}{k_{+2}(k_{+4}k_{+5} + k_{+4}k_{+6} + k_{+5}k_{+6})},$$

$$K_{mC}=\frac{k_{+5}k_{+6}(k_{-3}+k_{+4})}{k_{+3}(k_{+4}k_{+5}+k_{+4}k_{+6}+k_{+5}k_{+6})},\ K_{mP}=\frac{k_{-1}k_{-2}(k_{-3}+k_{+4})}{k_{-4}(k_{-1}k_{-2}+k_{-1}k_{-3}+k_{-2}k_{-3})},$$

$$K_{mQ}=\frac{k_{-1}k_{-2}k_{-3}}{k_{-5}(k_{-1}k_{-2}+k_{-1}k_{-3}+k_{-2}k_{-3})},\ K_{mR}=\frac{k_{-1}k_{-2}k_{-3}}{k_{-6}(k_{-1}k_{-2}+k_{-1}k_{-3}+k_{-2}k_{-3})},$$

$$K_{ia}=\frac{k_{-1}}{k_{+1}},\ K_{ib}=\frac{k_{-2}}{k_{+2}},\ K_{ic}=\frac{k_{-3}}{k_{+3}},\ K_{ip}=\frac{k_{+4}}{k_{-4}},\ K_{iq}=\frac{k_{+5}}{k_{-5}},\ K_{ir}=\frac{k_{+6}}{k_{-6}},$$

$$V_{mf}=[\mathrm{E}]_t\frac{k_{+4}k_{+5}k_{+6}}{k_{+4}k_{+5}+k_{+5}k_{+6}+k_{+4}k_{+6}},V_{mr}=[\mathrm{E}]_t\frac{k_{-1}k_{-2}k_{-3}}{k_{-1}k_{-2}+k_{-2}k_{-3}+k_{-1}k_{-3}}.$$

The kinetic parameters for this enzyme are obtained from Cha and Parks' data (8), at pH = 7.4 and $[Mg^{2+}]$ = 10 mM; these conditions are not ideally close to physiological. Kohn et al. account for the high $[Mg^{2+}]$ by assuming that GTP and GDP were essentially 100% chelated with magnesium in the experiments and use $MgGDP^-$ and $MgGTP^{2-}$ as species in their kinetic model. All other reactants are implicitly treated as sums of species.

Parameter Values for Succinyl-CoA Synthetase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Cha and Parks (8) (μM)	Model values (μM)
K_{mA}	8	8	2-8	16[a]
K_{mB}	660	55	11-55	55[b]
K_{mC}	55	660	150-660	660.0[b]
K_{mP}	335800	20	20	20
K_{mQ}	1600	880	400-800	880
K_{mR}	9.5	10	5-10	11.1[a]
K_{ia}	44.8	--	--	5.5[c]
K_{ib}	3700	--	100	100
K_{ic}	107.3	--	2000	2000
K_{ip}	4000	--	20	20
K_{iq}	2500	--	3000	3000
K_{ir}	15.3	--	10	11.1[a]

[a] These values have been corrected to apply to total reactant concentration for GDP and GTP (refer to the Isocitrate Dehydrogenase part).
[b] The parameter values computed from the KAG code result in switching the values of for K_{mB} and K_{mC} compared to the values published by Kohn et al. and Cha and Parks. We use the consistent published values and assume that the discrepancy is a bug in the code.

[c] This value computed from the Haldane constraint $K_{eq} = (K_{ip}K_{iq}K_{ir})/(K_{ia}K_{ib}K_{ic})$ (41) and using the apparent equilibrium constant of $K_{eq} = 0.61$ (2).

7. Succinate Dehydrogenase

The reference reaction is

$$\mathrm{SUC^{2-}} + \mathrm{COQ^0} \rightleftharpoons \mathrm{QH_2^0} + \mathrm{FUM^{2-}},$$

and the biochemical reaction is

$$\mathrm{SUC} + \mathrm{COQ} \rightleftharpoons \mathrm{QH_2} + \mathrm{FUM}.$$

The equilibrium constant for the reference reaction is:

$$K^0_{eq,\mathrm{sdh}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{sdh}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\mathrm{QH_2}} + \Delta_f G^0_{\mathrm{FUM}} - \Delta_f G^0_{\mathrm{SUC}} - \Delta_f G^0_{\mathrm{COQ}}}{RT}\right) = 1.69 \ .$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{sdh}} = K^0_{eq,\mathrm{sdh}} \frac{P_{\mathrm{QH_2}} P_{\mathrm{FUM}}}{P_{\mathrm{SUC}} P_{\mathrm{COQ}}} .$$

The reaction mechanism is assumed to be Theorell-Chance bi-bi when the reactant water is ignored (41), with the following fundamental steps (26):

$$\mathrm{E} + \mathrm{SUC^{2-}} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E \cdot SUC^{2-}}$$

$$\mathrm{E \cdot SUC^{2-}} + \mathrm{COQ^0} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \mathrm{E \cdot FUM^{2-}} + \mathrm{QH_2^0}$$

$$\mathrm{E \cdot FUM^{2-}} \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} \mathrm{E} + \mathrm{FUM^{2-}}.$$

The net reaction for this mechanism, in terms of chemical species is

$$\mathrm{SUC^{2-}} + \mathrm{COQ^0} \rightleftharpoons \mathrm{QH_2^0} + \mathrm{FUM^{2-}}.$$

In addition to the above mechanism, Kohn et al's argued that the oxaloacetate competitively inhibited the enzyme and succinate and fumarate competed with oxaloacetate for binding sites to "activate" the enzyme, and they modeled substrate (FUM^{2-} and SUC^{2-}) activation as irreversible displacement of an inhibitor (OAA^{2-}) from the inhibited enzyme (26). In our work, rapid equilibriums are assumed to occur between the activators and the free enzyme, which is consistent with a substrate activation mechanism proposed by Gutman (16).

The overall flux expression is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [SUC], [B] = [COQ], [P] = [QH_2], and [Q] = [FUM], we have

$$J_{\text{sdh}} = \frac{V_{mf}V_{mr}\left([\text{A}][\text{B}] - \dfrac{[\text{P}][\text{Q}]}{K_{eq,\text{sdh}}}\right)}{\begin{array}{r} V_{mr}K_{ia}K_{mB}\alpha_i + V_{mr}K_{mB}[\text{A}] + V_{mr}K_{mA}\alpha_i[\text{B}] + \dfrac{V_{mf}K_{mQ}\alpha_i}{K_{eq,\text{sdh}}}[\text{P}] + \dfrac{V_{mf}K_{mP}}{K_{eq,\text{sdh}}}[\text{Q}] \\ + V_{mr}[\text{A}][\text{B}] + \dfrac{V_{mf}K_{mQ}}{K_{eq,\text{sdh}}K_{ia}}[\text{A}][\text{P}] + \dfrac{V_{mr}K_{mA}}{K_{iq}}[\text{B}][\text{Q}] + \dfrac{V_{mf}}{K_{eq,\text{sdh}}}[\text{P}][\text{Q}] \end{array}},$$

where

$$K_{mA} = \frac{k_{+3}}{k_{+1}},\ K_{mB} = \frac{k_{+3}}{k_{+2}},\ K_{mP} = \frac{k_{-1}}{k_{-2}},\ K_{mQ} = \frac{k_{-1}}{k_{-3}},\ K_{ia} = \frac{k_{-1}}{k_{+1}},\ K_{ib} = \frac{k_{-1}}{k_{+2}},$$

$$K_{ip} = \frac{k_{+3}}{k_{-2}},\ K_{iq} = \frac{k_{+3}}{k_{-3}},\ V_{mf} = [\text{E}]_t k_{+3},\ V_{mr} = [\text{E}]_t k_{-1},$$

$$\alpha_i = \left(1 + \frac{[\text{OAA}]}{K_{i\text{OAA}}} + \frac{[\text{SUC}]}{K_{a\text{SUC}}} + \frac{[\text{FUM}]}{K_{a\text{FUM}}}\right) \Big/ \left(1 + \frac{[\text{SUC}]}{K_{a\text{SUC}}} + \frac{[\text{FUM}]}{K_{a\text{FUM}}}\right).$$

Parameter Values for Succinate Dehydrogenase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Hatefi and Stiggal (17) (μM)	Barman (3) (μM)	Gutman (16) (μM)	Model values (μM)
K_{mA}	46.7	467	300-1300	1300[c]	--	467
K_{mB}	119	480	480[b]	--	--	480
K_{mP}	2.45	2.45	--	--	--	2.45
K_{mQ}	1200	1200	--	1900[d]	--	1200
K_{ia}	5.14	120	--	1900[d]	110[e]	120
K_{iq}	10900	1275	1300	1300[d]	1300[e]	1275
$K_{i\text{OAA}}$	--	1.5	1.5[c]	--	--	1.5
$K_{a\text{SUC}}$	--	450[a]	--	--	800[f]	450
$K_{a\text{FUM}}$	--	375[a]	--	--	6400[f]	375

[a] Kohn et al. cited K_a values from Gopher and Gutman's abstract (15).
[b] This K_m value is obtained for phenazine methosulfate (PMS) used as electron acceptor.
[c] This value is measured from bovine heart mitochondria
[d] These values are measured from bovine heart with PMS as electron acceptor.
[e] These values are measured from pig heart and collected from Veeger and his collegue's work (11, 57).
[f] Gutman provided new set of K_a values with detailed discussion in his later publication (16). The K_a values are measured from bovine heart mitochondria.

8. Fumarase

The reference reaction is

$$\mathrm{FUM^{2-}} + \mathrm{H_2O} \rightleftharpoons \mathrm{MAL^{2-}},$$

and the biochemical reaction is

$$\mathrm{FUM} + \mathrm{H_2O} \rightleftharpoons \mathrm{MAL}.$$

The equilibrium constant for the reference reaction is:

$$K_{eq,\mathrm{fum}}^{0} = \exp\left(-\frac{\Delta_r G_{\mathrm{fum}}^{0}}{RT}\right) = \exp\left(-\frac{\Delta_f G_{\mathrm{MAL}}^{0} - \Delta_f G_{\mathrm{FUM}}^{0} - \Delta_f G_{\mathrm{H_2O}}^{0}}{RT}\right) = 4.04\ \mathrm{M^{-1}}.$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{fum}} = K_{eq,\mathrm{fum}}^{0}\frac{P_{\mathrm{MAL}}}{P_{\mathrm{FUM}}}.$$

The reaction mechanism is proposed to be ordered uni-uni when the reactant water is ignored (41), with the following fundamental steps (26):

$$\mathrm{E} + \mathrm{FUM^{2-}} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E \cdot FUM^{2-}}$$

$$\mathrm{E \cdot FUM^{2-}} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \mathrm{E} + \mathrm{MAL^{2-}}.$$

The net reaction for this mechanism, in terms of chemical species is

$$\mathrm{FUM^{2-}} \rightleftharpoons \mathrm{MAL^{2-}}.$$

In addition to the reaction mechanism, it is proposed that citrate, free ATP, free ADP, free GTP, and free GDP competitively inhibit fumarase.

The overall flux expression is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [FUM] and [P] = [MAL], we have

$$J_{\mathrm{fum}} = \frac{V_{mf}V_{mr}\left([\mathrm{A}] - \dfrac{[\mathrm{P}]}{K_{eq,\mathrm{fum}}}\right)}{K_{mA}V_{mr}\alpha_i + V_{mr}[\mathrm{A}] + \dfrac{V_{mf}[\mathrm{P}]}{K_{eq,\mathrm{fum}}}},$$

where

$$K_{mA} = \frac{k_{-1}+k_{+2}}{k_{+1}},\ K_{mP} = \frac{k_{-1}+k_{+2}}{k_{-2}},\ V_{mf} = [\mathrm{E}]_t k_{+3},\ V_{mr} = [\mathrm{E}]_t k_{-1},$$

$$\alpha_i = 1 + \frac{[\mathrm{CIT}]}{K_{i\mathrm{CIT}}} + \frac{[\mathrm{fATP}]}{K_{i\mathrm{ATP}}} + \frac{[\mathrm{fADP}]}{K_{i\mathrm{ADP}}} + \frac{[\mathrm{fGTP}]}{K_{i\mathrm{GTP}}} + \frac{[\mathrm{fGDP}]}{K_{i\mathrm{GDP}}}.$$

Parameter Values for Fumarase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Brant et al. (7) (μM)	Penner and Cohen (38) (μM)	Barman (3) (μM)	Model values (μM)
K_{mA}	44.7	44.7[a]	2.34[b]	--	1.74[d]	44.7
K_{mP}	193.5	197.7[a]	8[b]	--	3.79[d]	197.7
$K_{i\mathrm{CIT}}$	--	3500	--	--	3500[e]	3500
$K_{i\mathrm{ATP}}$	--	40	--	40[c]	--	40
$K_{i\mathrm{ADP}}$	--	400	--	400[c]	--	400
$K_{i\mathrm{GTP}}$	--	80	--	80[c]	--	80
$K_{i\mathrm{GDP}}$	--	330	--	330[c]	--	330

[a] These values have been corrected for prior protonation of fumarase and modified to satisfy Haldane relationship by Kohn et al., based on Brant et al.'s data.
[b] The enzyme used in Brant et al.'s experiments is prepared from pig hearts and these values are measured at 37 ℃.
[c] These values are the nucleotide concentrations required for 50% inhibition of yeast enzyme at 25 ℃ and pH = 7.0, and are cited by Kohn et al. as K_i values in their paper.
[d] These values are obtained from Brant et al.'s data and are measured at 21℃.
[e] This value is measured from pig heart at 23 ℃ and pH = 6.35.

9. Malate Dehydrogenase

The reference reaction is

$$\mathrm{NAD^+} + \mathrm{MAL^{2-}} \rightleftharpoons \mathrm{OAA^{2-}} + \mathrm{NADH^0} + \mathrm{H^+},$$

and the biochemical reaction is

$$\mathrm{NAD} + \mathrm{MAL} \rightleftharpoons \mathrm{OAA} + \mathrm{NADH}.$$

The equilibrium constant for the reference reaction is:

$$K^0_{eq,\mathrm{mdh}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{mdh}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\mathrm{OAA}} + \Delta_f G^0_{\mathrm{NADH}} - \Delta_f G^0_{\mathrm{NAD}} - \Delta_f G^0_{\mathrm{MAL}}}{RT}\right)$$

$$= 2.27\times 10^{-12} \ .$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{mdh}} = K^0_{eq,\mathrm{mdh}} \frac{1}{[\mathrm{H^+}]} \frac{P_{\mathrm{OAA}} P_{\mathrm{NADH}}}{P_{\mathrm{NAD}} P_{\mathrm{MAL}}}.$$

The reaction mechanism is proposed to be ordered bi-bi (41), with the following fundamental steps (26):

$$\mathrm{E} + \mathrm{NAD^+} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E}\cdot\mathrm{NAD^+}$$

$$\mathrm{E\cdot NAD^{+} + MAL^{2-}} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \mathrm{E\cdot NAD^{+}\cdot MAL^{2-}}$$

$$\mathrm{E\cdot NAD^{+}\cdot MAL^{2-}} \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} \mathrm{E\cdot NADH^{0} + OAA^{2-} + H^{+}}$$

$$\mathrm{E\cdot NADH^{0}} \underset{k_{-4}}{\overset{k_{+4}}{\rightleftharpoons}} \mathrm{E + NADH^{0}}.$$

The net reaction for this mechanism, in terms of chemical species is

$$\mathrm{NAD^{+} + MAL^{2-} \rightleftharpoons OAA^{2-} + H^{+} + NADH^{0}}.$$

In addition to the above reaction mechanism, it is proposed that adenine nucleotides are competitive inhibitors against NADH and NAD in the reaction (29).

The overall flux expression is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [NAD], [B] = [MAL], [P] = [OAA], and [Q] = [NADH], we have

$$J_{\mathrm{mdh}} = \frac{V_{mf}V_{mr}\left([\mathrm{A}][\mathrm{B}] - \frac{[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{mdh}}}\right)}{\begin{array}{l} V_{mr}K_{ia}K_{mB}\alpha_i + V_{mr}K_{mB}[\mathrm{A}] + V_{mr}K_{mA}\alpha_i[\mathrm{B}] + \frac{V_{mf}K_{mQ}\alpha_i[\mathrm{P}]}{K_{eq,\mathrm{mdh}}} + \frac{V_{mf}K_{mP}[\mathrm{Q}]}{K_{eq,\mathrm{mdh}}} \\ + V_{mr}[\mathrm{A}][\mathrm{B}] + \frac{V_{mf}K_{mQ}[\mathrm{A}][\mathrm{P}]}{K_{eq,\mathrm{mdh}}K_{ia}} + \frac{V_{mf}[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{mdh}}} + \frac{V_{mr}K_{mA}[\mathrm{B}][\mathrm{Q}]}{K_{iq}} \\ + \frac{V_{mr}[\mathrm{A}][\mathrm{B}][\mathrm{P}]}{K_{ip}} + \frac{V_{mf}[\mathrm{B}][\mathrm{P}][\mathrm{Q}]}{K_{ib}K_{eq,\mathrm{mdh}}} \end{array}},$$

where

$$K_{mA} = \frac{k_{+3}k_{+4}}{k_{+1}(k_{+3}+k_{+4})},\ K_{mB} = \frac{k_{+4}(k_{-2}+k_{+3})}{k_{+2}(k_{+3}+k_{+4})},\ K_{ia} = \frac{k_{-1}}{k_{+1}},\ K_{ib} = \frac{k_{-1}+k_{-2}}{k_{+2}},$$

$$K_{mP} = \frac{k_{-1}(k_{-2}+k_{+3})}{k_{-3}(k_{-1}+k_{-2})},\ K_{mQ} = \frac{k_{-1}k_{-2}}{k_{-4}(k_{-1}+k_{-2})},\ K_{ip} = \frac{k_{+3}+k_{+4}}{k_{-3}},\ K_{iq} = \frac{k_{+4}}{k_{-4}}$$

$$V_{mf} = [\mathrm{E}]_t\frac{k_{+3}k_{+4}}{k_{+3}+k_{+4}},\ V_{mr} = [\mathrm{E}]_t\frac{k_{-1}k_{-2}}{k_{-1}+k_{-2}},$$

$$\alpha_i = 1 + \frac{[\mathrm{fATP}]}{K_{i\mathrm{ATP}}} + \frac{[\mathrm{fADP}]}{K_{i\mathrm{ADP}}} + \frac{[\mathrm{fAMP}]}{K_{i\mathrm{AMP}}}.$$

Parameter Values for Malate Dehydrogenase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM	Kohn et al. (26, 29) (μM)	Kimball et al. (22) (μM)	Oza and Shore (36) (μM)	Model values (μM)

	codes (24) (μM)				
K_{mA}	90.56	90.55[a]	300[c]	--	90.55
K_{mB}	250	250[a]	800[c]	--	250
K_{mP}	32.2	6.128[a]	40[c]	--	6.128
K_{mQ}	242500	2.58[a]	16[c]	--	2.58
K_{ia}	279	279[a]	900[c]	--	279
K_{ib}	621	360[a]	--	--	360
K_{ip}	49.9	5.5[a]	--	--	5.5
K_{iq}	299000	3.18[a]	5[c]	--	3.18
$K_{i\mathrm{ATP}}$	--	183.2[b]	--	709.3[d]	183.2
$K_{i\mathrm{ADP}}$	--	394.4[b]	--	383.2[d]	394.4
$K_{i\mathrm{AMP}}$	--	420.0[b]	--	793.0[d]	420.0

[a] These values are adjusted by Kohn and Garfinkel to satisfy Haldane relationship.
[b] These values are calculated for ATP^{4-}, ADP^{3-}, and AMP^{2-} by Kohn and Garfinkel, based on Oza and Shore's data.
[c] These values are measured from pig heart at 25℃ and pH = 8.0.
[d] These values are computed for unchelated ATP, ADP, and AMP by the authors from Oza and Shore's data, using the relationship $\frac{K_{m,app}}{K_m} = 1 + \frac{[I]}{K_{iI}}$ (41, 43), where [I] and K_{iI} denote inhibitor concentration and inhibition constant, respectively. Oza and Shore's data are obtained from pig heart mitochondria.

10. Nucleoside Diphosphokinase

The reference reaction is

$$\mathrm{GTP}^{4-} + \mathrm{ADP}^{3-} \rightleftharpoons \mathrm{GDP}^{3-} + \mathrm{ATP}^{4-},$$

and the biochemical reaction is

$$\mathrm{GTP} + \mathrm{ADP} \rightleftharpoons \mathrm{GDP} + \mathrm{ATP}\ .$$

Adenosine and guanosine di- and tri-phosphates are usually regarded as energetically equivalent pairs (35). The equilibrium constant for the reference reaction is:

$$K^0_{eq,\mathrm{ndk}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{ndk}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\mathrm{GDP}} + \Delta_f G^0_{\mathrm{ATP}} - \Delta_f G^0_{\mathrm{GTP}} - \Delta_f G^0_{\mathrm{ADP}}}{RT}\right) = 1\,.$$

The apparent equilibrium constant for the biochemical reaction is equal to unity, i.e.,

$$K_{eq,\mathrm{nkd}} = K^0_{eq,\mathrm{nkd}} = 1\,.$$

The reaction mechanism is proposed to be ping-pong-bi-bi (41), with the following fundamental steps (26, 37):

$$\mathrm{E} + \mathrm{MgGTP}^{2-} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E}\cdot\mathrm{MgGTP}^{2-}$$

$$\mathrm{E \cdot MgGTP^{2-}} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \mathrm{F + MgGDP^{-}}$$

$$\mathrm{F + MgADP^{-}} \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} \mathrm{E \cdot MgATP^{2-}}$$

$$\mathrm{E \cdot MgATP^{2-}} \underset{k_{-4}}{\overset{k_{+4}}{\rightleftharpoons}} \mathrm{E + MgATP^{2-}}.$$

The net reaction for this mechanism, in terms of chemical species is

$$\mathrm{MgGTP^{2-} + MgADP^{-} \rightleftharpoons MgATP^{2-} + MgGDP^{-}}.$$

In addition to the above reaction mechanism, it is proposed that AMP is a noncompetitive inhibitor for the reaction (26).

The overall flux expression is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [GTP], [B] = [ADP], [P] = [GDP], and [Q] = [ATP], we have

$$J_{\mathrm{ndk}} = \frac{V_{mf}V_{mr}\left([\mathrm{A}][\mathrm{B}] - \frac{[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{ndk}}}\right)\Big/\alpha_i}{V_{mr}K_{mB}[\mathrm{A}] + V_{mr}K_{mA}[\mathrm{B}] + \frac{V_{mf}K_{mQ}[\mathrm{P}]}{K_{eq,\mathrm{ndk}}} + \frac{V_{mf}K_{mP}[\mathrm{Q}]}{K_{eq,\mathrm{ndk}}} + V_{mr}[\mathrm{A}][\mathrm{B}] + \frac{V_{mf}K_{mQ}[\mathrm{A}][\mathrm{P}]}{K_{eq,\mathrm{ndk}}K_{ia}} + \frac{V_{mf}[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{ndk}}} + \frac{V_{mr}K_{mA}[\mathrm{B}][\mathrm{Q}]}{K_{iq}}},$$

where

$$K_{mA} = \frac{k_{+3}k_{+4}(k_{+2}+k_{-1})}{k_{+1}k_{+3}(k_{+2}+k_{+4})},\ K_{mB} = \frac{k_{+2}(k_{-3}+k_{+4})}{k_{+3}(k_{+2}+k_{+4})},\ K_{ia} = \frac{k_{-1}}{k_{+1}},\ K_{ib} = \frac{k_{-3}}{k_{+3}},\ K_{mP} = \frac{k_{-3}(k_{-1}+k_{+2})}{k_{-2}(k_{-1}+k_{-3})},$$

$$K_{mQ} = \frac{k_{-1}(k_{-3}+k_{+4})}{k_{-4}(k_{-1}+k_{-3})},\ K_{ip} = \frac{k_{+2}}{k_{-2}},\ K_{iq} = \frac{k_{+4}}{k_{-4}},\ V_{mf} = [\mathrm{E}]_t\frac{k_{+2}k_{+4}}{k_{+2}+k_{+4}},\ V_{mr} = [\mathrm{E}]_t\frac{k_{-1}k_{-3}}{k_{-1}+k_{-3}},$$

$$\alpha_i = 1 + \frac{[\mathrm{fAMP}]}{K_{i\mathrm{AMP}}}.$$

Parameter Values for Nucleoside Diphosphokinase

Parameter	Computed from rate constants in KAG 1981 BiOSSIM codes (24) (μM)	Kohn et al. (26, 29) (μM)	Parks and Agarwal (37) (μM)	Barman (3) (μM)	Model values (μM)
K_{mA}	100	100[a]	50-200[b]	--	111[e]
K_{mB}	50	50[a]	15-1500[b]	40[d]	100[e]
K_{mP}	130	130[a]	130[c]	--	260[e]

K_{mQ}	250	250[a]	80-8000[b]	--	278[e]
K_{ia}	143.7	--	--	--	170[e]
K_{ib}	71.8	--	--	--	143.6[e]
K_{ip}	73.3	--	--	--	146.6[e]
K_{iq}	140.9	--	--	--	156.5[e]
$K_{i\mathrm{AMP}}$	--	650	--	650[d]	650

[a] These values are assumed to be Michaelis constants of magnesium-bound nucleotides.
[b] These K_m values are obtained from various sources, such as human erythrocytes, yeast, *B. subtilis*, calf liver, calf thymus, beef heart, and rat liver.
[c] This value is determined for dGDP (Deoxyguanosine diphosphate**)** at pH = 5.8 with the enzyme extracted from human erythrocytes.
[d] These values are measured from human erythrocyte and at 30℃ and pH = 7.5. The K_i of GMP is used here for AMP.
[e] These values have been modified to apply to total reactant concentration for nucleotides. (See explanation under Isocitrate Dehyrogenase)

11. Glutamate Oxaloacetate Transaminase (Aspartate Transaminase)

The reference reaction is

$$\mathrm{ASP^{-}} + \mathrm{AKG^{2-}} \rightleftharpoons \mathrm{OAA^{2-}} + \mathrm{GLU^{-}},$$

and the biochemical reaction is

$$\mathrm{ASP} + \mathrm{AKG} \rightleftharpoons \mathrm{OAA} + \mathrm{GLU}\,.$$

The equilibrium constant for the reference reaction is:

$$K^0_{eq,\mathrm{got}} = \exp\left(-\frac{\Delta_r G^0_{\mathrm{got}}}{RT}\right) = \exp\left(-\frac{\Delta_f G^0_{\mathrm{OAA}} + \Delta_f G^0_{\mathrm{GLU}} - \Delta_f G^0_{\mathrm{ASP}} - \Delta_f G^0_{\mathrm{AKG}}}{RT}\right) = 1.77\,.$$

The apparent equilibrium constant for the biochemical reaction is

$$K_{eq,\mathrm{got}} = K^0_{eq,\mathrm{got}} \frac{P_{\mathrm{OAA}} P_{\mathrm{GLU}}}{P_{\mathrm{ASP}} P_{\mathrm{AKG}}}.$$

The reaction was proposed to be ping-pong-bi-bi with the following steps (18, 24, 41):

$$\mathrm{E} + \mathrm{ASP^{-}} \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} \mathrm{E\cdot ASP^{-}}$$

$$\mathrm{E\cdot ASP^{-}} \underset{k_{-2}}{\overset{k_{+2}}{\rightleftharpoons}} \mathrm{F} + \mathrm{OAA^{2-}}$$

$$\mathrm{F} + \mathrm{AKG^{2-}} \underset{k_{-3}}{\overset{k_{+3}}{\rightleftharpoons}} \mathrm{E\cdot GLU^{-}}$$

$$\mathrm{E\cdot GLU^{-}} \underset{k-4}{\overset{k_{+4}}{\rightleftharpoons}} \mathrm{E} + \mathrm{GLU^{-}}.$$

The net reaction for this mechanism, in terms of chemical species is

$$ASP^{-} + AKG^{2-} \rightleftharpoons OAA^{2-} + GLU^{-}.$$

In addition to the reaction mechanism, it is reported that high concentration α-ketoglutarate competitively inhibits aspartate (18).

The overall flux expression is derived from steady-state mass balance. Denoting the reactant concentrations as [A] = [ASP], [B] = [AKG], [P] = [OAA], and [Q] = [GLU], we have

$$J_{\mathrm{got}} = \frac{V_{mf}V_{mr}\left([\mathrm{A}][\mathrm{B}] - \dfrac{[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{got}}}\right)}{\begin{array}{l} V_{mr}K_{mB}[\mathrm{A}] + V_{mr}K_{mA}\alpha_i[\mathrm{B}] + \dfrac{V_{mf}K_{mQ}\alpha_i[\mathrm{P}]}{K_{eq,\mathrm{got}}} + \dfrac{V_{mf}K_{mP}[\mathrm{Q}]}{K_{eq,\mathrm{got}}} + V_{mr}[\mathrm{A}][\mathrm{B}] \\ + \dfrac{V_{mf}K_{mQ}[\mathrm{A}][\mathrm{P}]}{K_{eq,\mathrm{got}}K_{ia}} + \dfrac{V_{mf}[\mathrm{P}][\mathrm{Q}]}{K_{eq,\mathrm{got}}} + \dfrac{V_{mr}K_{mA}[\mathrm{B}][\mathrm{Q}]}{K_{iq}} \end{array}},$$

where

$$K_{mA} = \frac{k_{+3}k_{+4}(k_{+2}+k_{-1})}{k_{+1}k_{+3}(k_{+2}+k_{+4})},\ K_{mB} = \frac{k_{+2}(k_{-3}+k_{+4})}{k_{+3}(k_{+2}+k_{+4})},\ K_{ia} = \frac{k_{-1}}{k_{+1}},\ K_{ib} = \frac{k_{-3}}{k_{+3}},\ K_{mP} = \frac{k_{-3}(k_{-1}+k_{+2})}{k_{-2}(k_{-1}+k_{-3})},$$

$$K_{mQ} = \frac{k_{-1}(k_{-3}+k_{+4})}{k_{-4}(k_{-1}+k_{-3})},\ K_{ip} = \frac{k_{+2}}{k_{-2}},\ K_{iq} = \frac{k_{+4}}{k_{-4}},\ V_{mf} = [\mathrm{E}]_t\frac{k_{+2}k_{+4}}{k_{+2}+k_{+4}},\ V_{mr} = [\mathrm{E}]_t\frac{k_{-1}k_{-3}}{k_{-1}+k_{-3}},$$

$$\alpha_i = 1 + \frac{[\mathrm{AKG}]}{K_{i\mathrm{AKG}}}.$$

Parameter Values for Glutamate Oxaloacetate Transaminase

Parameter	Computed from rate constants BiOSSIM codes (24) (μM)	Henson and Cleland (18) (μM) [b]	Model values (μM)
K_{mA}	4800	3900	3900
K_{mB}	418	430	430
K_{mP}	861500	88	88
K_{mQ}	113000[a]	8900	8900
K_{ia}	4400	3480	3480
K_{ib}	804	710	710
K_{ip}	50.9	50	50
K_{iq}	135900[a]	8400	8400
$K_{i\mathrm{AKG}}$	--	16600	16600

[a] These values are adjusted by Kohn and Garfinkel to satisfy Haldane relationship.
[b] These values are measured from pig heart mitochondria, at 37 ℃ and pH = 7.4.

Appendix D: The Differential Equations for pH, Mg^{2+}, and K^+

The differential equations for pH and binding are determined based on mass balance. The kinetics of pH is governed by proton binding and unbinding as well as the consumption and generation of protons via chemical reactions. For a general system of N_r reactants, and considering [H^+], [Mg^{2+}] and [K^+] binding, the total concentration of protons bound to reactants is calculated

$$[\mathrm{H}_{bound}]=\sum_{i=1}^{N_r}\left([\mathrm{L}_i\mathrm{H}_1]+2[\mathrm{L}_i\mathrm{H}_2]+...\right), \tag{D1}$$

where i indexes the reactants denoted by L_i; the concentrations of the one- and two-proton bound species of reactant i are denoted by [L_iH_1] and [L_iH_2], respectively. If we ignore species with two or more protons bound, then

$$[\mathrm{H}_{bound}] = \sum_{i=1}^{\mathrm{N}_r}[\mathrm{L}_i\mathrm{H}_1] = \sum_{i=1}^{\mathrm{N}_r}[\mathrm{L}_i]\frac{[\mathrm{H}^+]/K_i^H}{P_i\left([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+]\right)}, \tag{D2}$$

where [L_i] is the total concentration for reactant i, K_i^H and P_i are the proton binding constant and binding polynomial for the reactant. The binding polynomials are calculated

as $P_i\left([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+]\right) = 1+[\mathrm{H}^+]/K_i^H+[\mathrm{Mg}^{2+}]/K_i^{Mg}+[\mathrm{K}^+]/K_i^K$, (D3)

where K_i^{Mg} and K_i^K are the binding constants for Mg^{2+} and K^+. Equation (D2) expresses the total concentration of reversibly bound protons as a function of the total reactant concentrations in the system, and the pH and free concentrations of Mg^{2+} and K^+. The rate of change of free [H^+], [Mg^{2+}] and [K^+] can be calculated based on mass conservation. The some of the terms in the following equations are defined and explained subsequently in this section.

$$J_t^H = \frac{d[\mathrm{H}^+]}{dt}+\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{H}^+]}\frac{d[\mathrm{H}^+]}{dt}+\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\frac{d[\mathrm{Mg}^{2+}]}{dt}+\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]}\frac{d[\mathrm{K}^+]}{dt}+\sum_{k=1}^{N_f}n_kJ_k \tag{D4}$$

$$J_t^{Mg} = \frac{d[\mathrm{Mg}^{2+}]}{dt}+\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]}\frac{d[\mathrm{Mg}^{2+}]}{dt}+\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\frac{d[\mathrm{Mg}^{2+}]}{dt}+\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]}\frac{d[\mathrm{K}^+]}{dt} \tag{D5}$$

$$J_t^K = \frac{d[\mathrm{K}^+]}{dt}+\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]}\frac{d[\mathrm{H}^+]}{dt}+\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\frac{d[\mathrm{Mg}^{2+}]}{dt}+\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{K}^+]}\frac{d[\mathrm{K}^+]}{dt} \tag{D6}$$

After algebraic manipulation, we are able to calculate time derivatives of [H^+], [Mg^{2+}], and [K^+]. But before giving explicit expressions for d[H^+]/dt, d[Mg^{2+}]/dt, and d[K^+]/dt, we write down expressions for partial derivatives, flux terms, and buffering terms first.

The partial derivatives are expressed

$$\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} = -\sum_{i=1}^{N_r}\frac{[\mathrm{L}_i][\mathrm{H}^+]/K_i^H}{K_i^{Mg}\left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D7}$$

$$\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]} = -\sum_{i=1}^{N_r}\frac{[\mathrm{L}_i][\mathrm{H}^+]/K_i^H}{K_i^K\left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D8}$$

$$\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{H}^+]} = \sum_{i=1}^{N_r} \frac{[\mathrm{L}_i]\left(1+[\mathrm{Mg}^{2+}]/K_i^{Mg} + [\mathrm{K}^+]/K_i^K\right)}{K_i^H \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D9}$$

$$\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]} = -\sum_{i=1}^{N_r} \frac{[\mathrm{L}_i][\mathrm{Mg}^{2+}]/K_i^{Mg}}{K_i^H \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D10}$$

(D11)

$$\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]} = -\sum_{i=1}^{N_r} \frac{[\mathrm{L}_i][\mathrm{Mg}^{2+}]/K_i^{Mg}}{K_i^K \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D12}$$

$$\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} = \sum_{i=1}^{N_r} \frac{[\mathrm{L}_i]\left(1+[\mathrm{H}^+]/K_i^{H} + [\mathrm{K}^+]/K_i^K\right)}{K_i^{Mg} \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D13}$$

$$\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]} = -\sum_{i=1}^{N_r} \frac{[\mathrm{L}_i][\mathrm{K}^{+}]/K_i^{K}}{K_i^H \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D14}$$

$$\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} = -\sum_{i=1}^{N_r} \frac{[\mathrm{L}_i][\mathrm{K}^{+}]/K_i^{K}}{K_i^{Mg} \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}, \tag{D15}$$

$$\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{K}^+]} = \sum_{i=1}^{N_r} \frac{[\mathrm{L}_i]\left(1+[\mathrm{H}^+]/K_i^{H} + [\mathrm{Mg}^{2+}]/K_i^{Mg}\right)}{K_i^{K} \left(P_i([\mathrm{H}^+],[\mathrm{Mg}^{2+}],[\mathrm{K}^+])\right)^2}. \tag{D16}$$

The flux terms for H^+, Mg^{2+}, and K^+ are:

$$\Phi^H = -\sum_{i=1}^{N_r} \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{L}_i]} \frac{d[\mathrm{L}_i]}{dt} + \sum_{k=1}^{N_f} n_k J_k + J_t^H, \tag{D17}$$

$$\Phi^{Mg} = -\sum_{i=1}^{N_r} \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{L}_i]} \frac{d[\mathrm{L}_i]}{dt} + J_t^{Mg}, \tag{D18}$$

$$\Phi^{K} = -\sum_{i=1}^{N_r} \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{L}_i]} \frac{d[\mathrm{L}_i]}{dt} + J_t^{K}, \tag{D19}$$

where N_r is the number of reactants, N_f is the number of reactions, n_k is the stoichiometric coefficient of k^{th} reaction, J_k is the flux of k^{th} reaction, J_t^H (J_t^{Mg}, J_t^K) is the transport flux of $[\mathrm{H}^+]$ ($[\mathrm{Mg}^{2+}]$, $[\mathrm{K}^+]$) into the system. In the current model, for the mitochondrial matrix, we have

$$\sum_{k=1}^{N_f} n_k J_k = \left(-J_{\mathrm{pdh}} + 2J_{\mathrm{cits}} - J_{\mathrm{akgd}} + J_{\mathrm{scoas}} + J_{\mathrm{mdh}}\right)/W_{\mathrm{x}}, \tag{D20}$$

$$\begin{aligned} J_t^H = \big(&J_{\mathrm{PYRH}} + J_{\mathrm{GLUH}} + J_{\mathrm{CITMAL}} - J_{\mathrm{ASPGLU}} - 5J_{\mathrm{C1}} - 2J_{\mathrm{C3}} - 4J_{\mathrm{C4}} + (n_A - 1)J_{\mathrm{F1}} \\ &+ 2J_{\mathrm{PIHt}} + J_{\mathrm{Hle}} - J_{\mathrm{KH}}\big)/W_{\mathrm{x}}, \end{aligned} \tag{D21}$$

$$J_t^{Mg} = 0, \tag{D22}$$

$$J_t^K = J_{\mathrm{KH}}/W_{\mathrm{x}}. \tag{D23}$$

The buffering terms are:

$$\alpha_H = 1 + \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{H}^+]} + \frac{B_x}{K_{Bx}\left(1+[\mathrm{H}^+]/K_{Bx}\right)^2}, \tag{D24}$$

$$\alpha_{Mg} = 1 + \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}, \tag{D25}$$

$$\alpha_K = 1 + \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{K}^+]}, \tag{D26}$$

where we have added an additional H^+-buffering term to account for buffering by biochemical species not explicitly accounted for in the model. The constants B_x and K_{Bx} are set to 0.02 M and 10^{-7} M, respectively, so that the computed matrix buffering capacity could be consistent with experimental values (21, 49) .

Then we compute the time derivatives of $[H^+]$, $[Mg^{2+}]$, and $[K^+]$ by using Equations (D27), (D28), and (D29):

$$\begin{aligned}\frac{d[\mathrm{H}^+]}{dt} = \Bigg[&\left(\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\cdot\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]} - \alpha_{Mg}\alpha_K\right)\Phi^H \\ &+ \left(\alpha_K\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} - \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]}\cdot\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\right)\Phi^{Mg} \\ &+ \left(\alpha_{Mg}\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]} - \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\cdot\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]}\right)\Phi^K\Bigg]/D,\end{aligned} \tag{D27}$$

$$\begin{aligned}\frac{d[\mathrm{Mg}^{2+}]}{dt} = \Bigg[&\left(\alpha_K\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]} - \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]}\cdot\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]}\right)\Phi^H \\ &+ \left(\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]}\cdot\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]} - \alpha_H\alpha_K\right)\Phi^{Mg} \\ &+ \left(\alpha_H\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]} - \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]}\cdot\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]}\right)\Phi^K\Bigg]/D,\end{aligned} \tag{D28}$$

$$\begin{aligned}\frac{d[\mathrm{K}^+]}{dt} = \Bigg[&\left(\alpha_{Mg}\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]} - \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\cdot\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]}\right)\Phi^H \\ &+ \left(\alpha_H\frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} - \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]}\cdot\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]}\right)\Phi^{Mg} \\ &+ \left(\frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]}\cdot\frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} - \alpha_H\alpha_{Mg}\right)\Phi^K\Bigg]/D,\end{aligned} \tag{D29}$$

where

$$
\begin{aligned}
D \quad = \;& \alpha_H \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} \cdot \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]} \; + \; \alpha_K \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} \cdot \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]} \\
& + \alpha_{Mg} \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]} \cdot \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]} \quad - \; \alpha_{Mg}\alpha_K\alpha_H \\
& - \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{K}^+]} \cdot \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} \cdot \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{H}^+]} \\
& - \frac{\partial[\mathrm{H}_{bound}]}{\partial[\mathrm{Mg}^{2+}]} \cdot \frac{\partial[\mathrm{Mg}_{bound}]}{\partial[\mathrm{K}^+]} \cdot \frac{\partial[\mathrm{K}_{bound}]}{\partial[\mathrm{H}^+]} .
\end{aligned}
\tag{D30}
$$

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